

Cyberphysical Systems: Part 2

Sayan Mitra

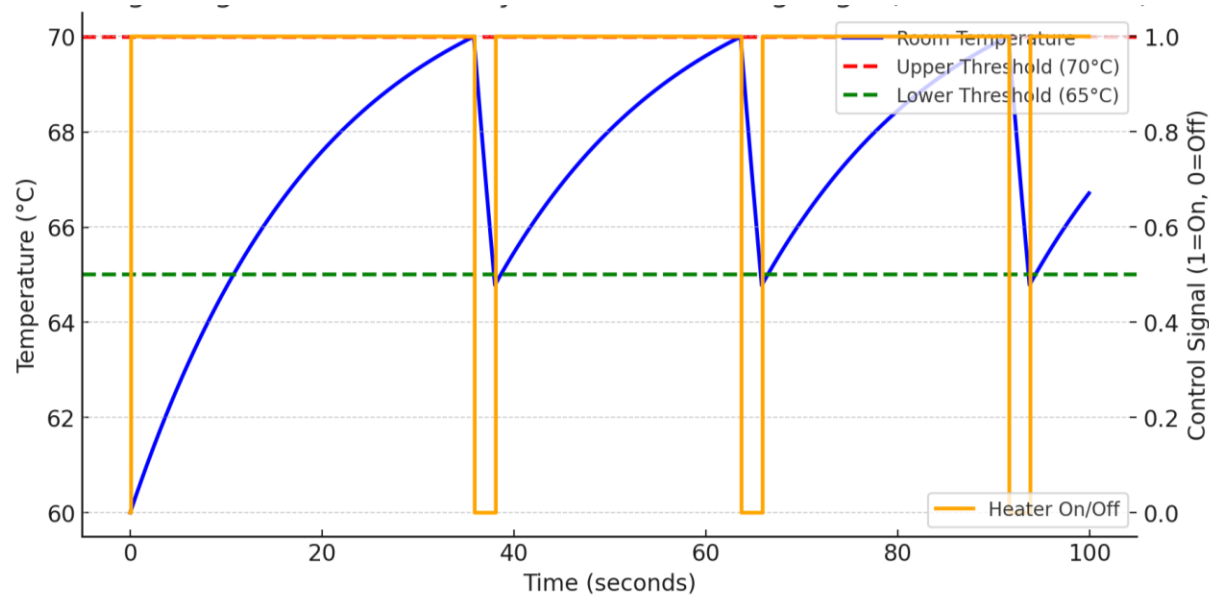
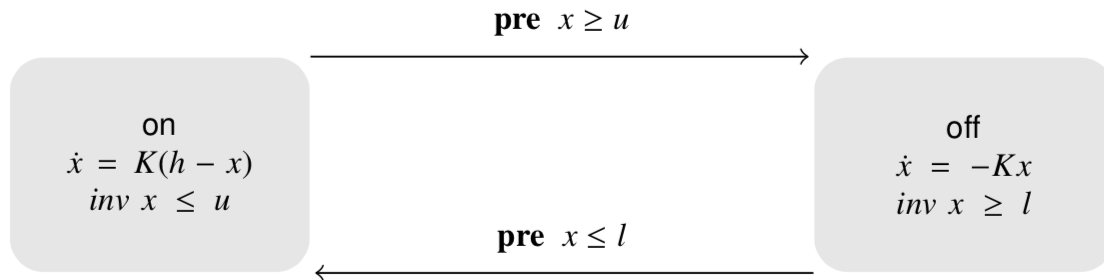
Verifying cyberphysical systems

mitras@illinois.edu

Outline

- Review: Modeling cyber-physical systems
 - Hybrid automata
 - Executions
 - Urgency
 - Zeno
 - Hybrid stability

Example. Thermostat automaton



automaton Thermostat($u, l, K, h : \text{Real}$) where $u > l$

type Status enumeration [on, off]

actions

turnOn; turnOff;

variables

$x : \text{Real} := l$ loc: Status := on

transitions

turnOn

pre $x \leq l \wedge \text{loc} = \text{off}$

eff loc := on

trajectories

modeOn

evolve $d(x) = K(h - x)$

invariant loc = on $\wedge x \leq u$

turnOff

pre $x \geq u \wedge \text{loc} = \text{on}$

eff loc := off

modeOff

evolve $d(x) = -Kx$

invariant loc = off $\wedge x \geq l$

Determinism vs nondeterminism

mode invariants

Hybrid Automaton

$$\mathcal{A} = (X, \Theta, A, \mathcal{D}, \mathcal{T})$$

- X : set of **state variables**
 - $Q \subseteq \text{val}(X)$ set of **states**
- $\Theta \subseteq Q$ set of **start states**
- set of **actions**, $A = E \cup H$
- $\mathcal{D} \subseteq Q \times A \times Q$
- $\mathcal{T}$: set of **trajectories** for X which is closed under **prefix, suffix, and concatenation**

Semantics: Executions and Traces

An **execution** of $\mathcal{A}$ is an (possibly infinite) alternating sequence $\alpha = \tau_0 a_1 \tau_1 a_2 \tau_2 \dots$ where

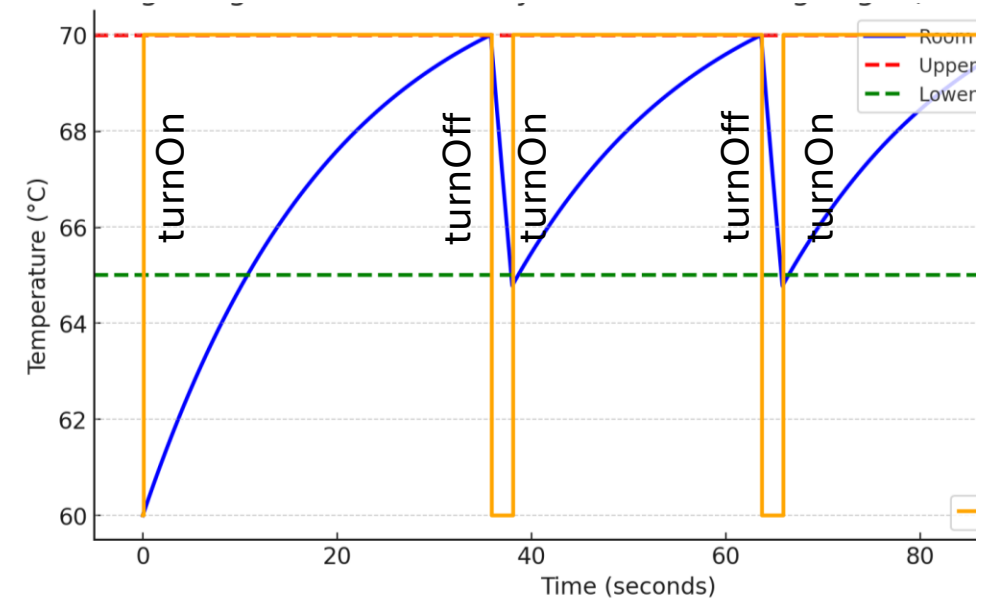
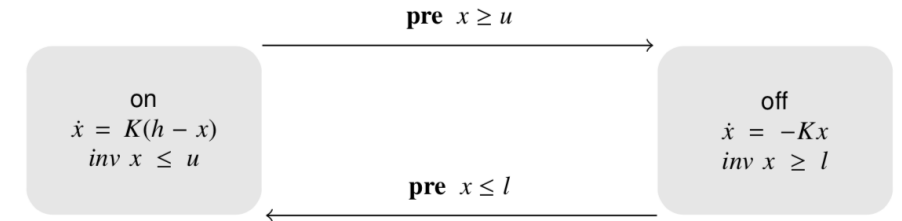
- $\forall i, \tau_i.lstate \xrightarrow{a_{i+1}} \tau_{i+1}.fstate$
- If $\tau_0.fstate \in \Theta$ then α is an **execution**

Execs $_{\mathcal{A}}$ set of all executions

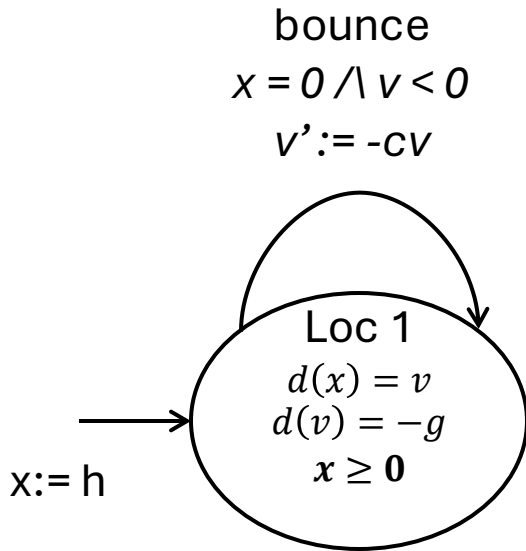
First state of α is $\alpha.fstate = \tau_0.fstate$

If α is **finite and closed** $\tau_0 a_1 \tau_1 a_2 \tau_2 \dots \tau_k$ then $\alpha.lstate = \tau_k.lstate$

A state x is *reachable* if there exists an execution α with $\alpha.lstate = x$



Example. Bouncing Ball



Graphical Representation used in many articles

automaton Bouncingball(c,h,g)

variables: x : Reals := h, v : Reals := 0

actions: bounce

transitions:

bounce

pre $x = 0 \wedge v < 0$

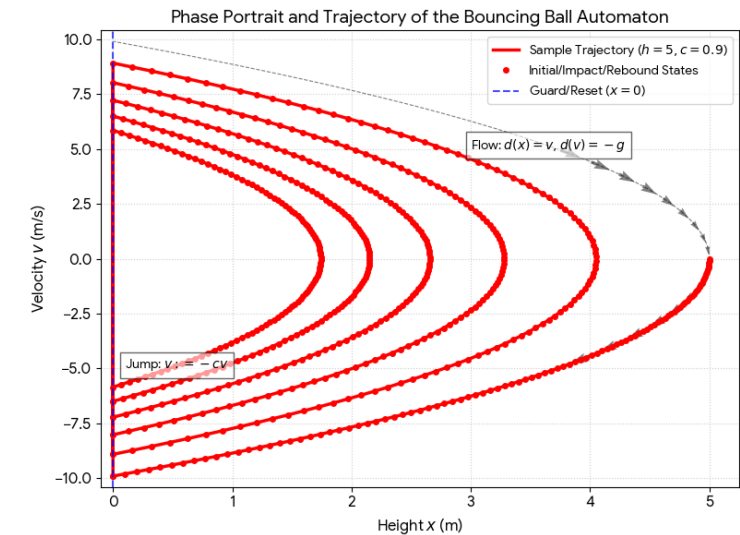
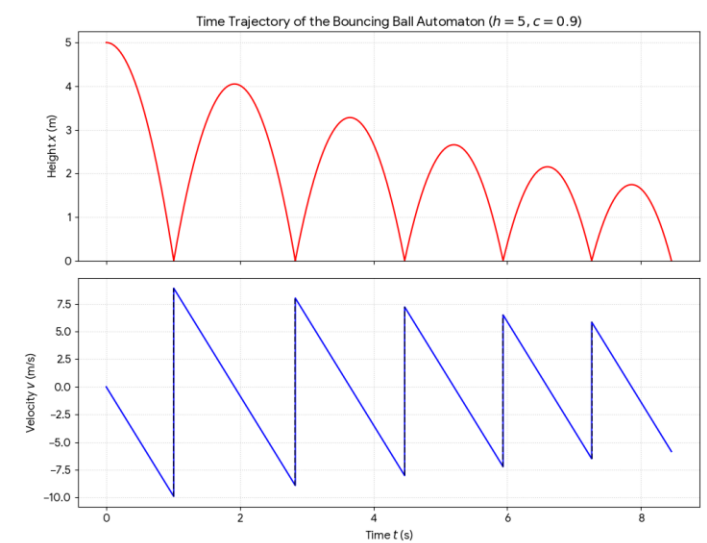
eff $v := -cv$

trajectories:

Loc1

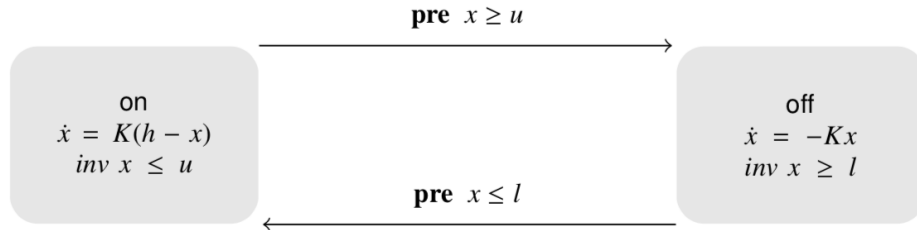
evolve $d(x) = v; d(v) = -g$

invariant $x \geq 0$



mode invariant, not to be confused with invariants of the automaton

Urgency: Must transitions



An **urgent** transition (or action) is an action that has to occur as soon as it is enabled

Introduce special syntax for creating urgent transitions, but we can use mode invariants to accomplish this

automaton Thermostat($u, l, K, h, d : \text{Real}$) where $u > l$

type *Status* enumeration [*on*, *off*]

actions

turnOn; turnOff;

variables

$x : \text{Real} := l$ $loc : \text{Status} := on$

transitions

turnOn

pre $x \leq l \wedge loc = off$

eff $loc := on$

turnOff

pre $x \geq u \wedge loc = on$

eff $loc := off$

trajectories

modeOn

evolve $d(x) = K(h - x)$

invariant $loc = on \wedge x \leq u + d$

modeOff

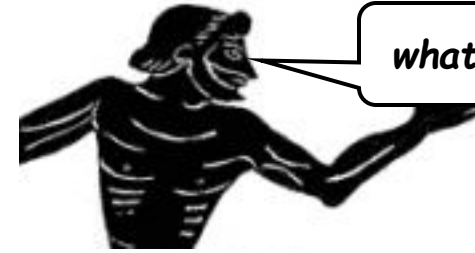
evolve $d(x) = -Kx$

invariant $loc = off \wedge x \geq l - d$

Zeno's Paradox

Achilles, the fastest athlete, greatest warrior

Zeno, Greek philosopher

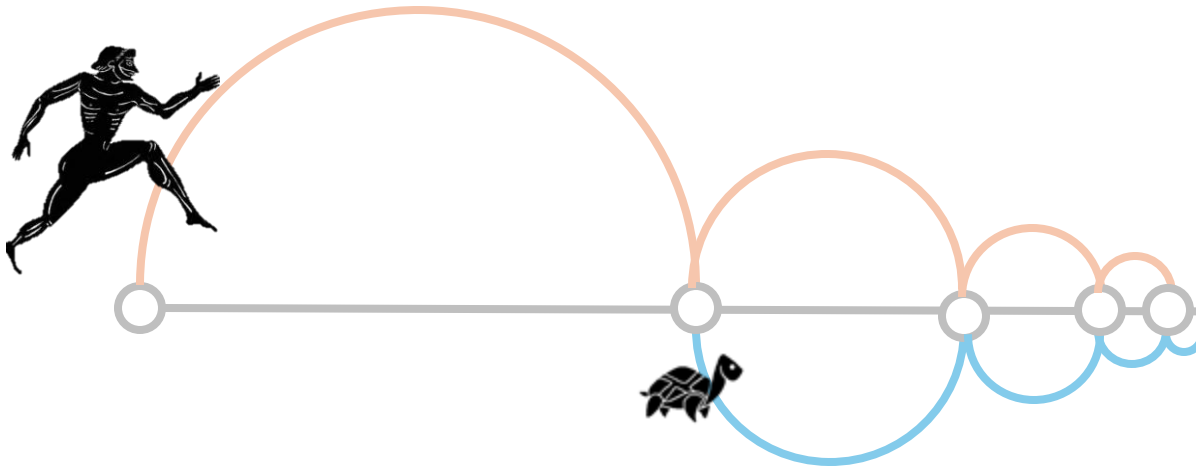


whatever!

You couldn't even beat a turtle



Achilles runs 10 times faster than the tortoise, but the turtle gets to start 1 second earlier. Can Achilles ever catch Turtle?



After $1/10^{\text{th}}$ of a second, Achilles reaches where the Turtle (T) started, and T has a head start of $1/10^{\text{th}}$ second.

After another $1/100^{\text{th}}$ of a second, A catches up to where T was at $t=1/10$ sec, but T has a head start of $1/100^{\text{th}}$

...

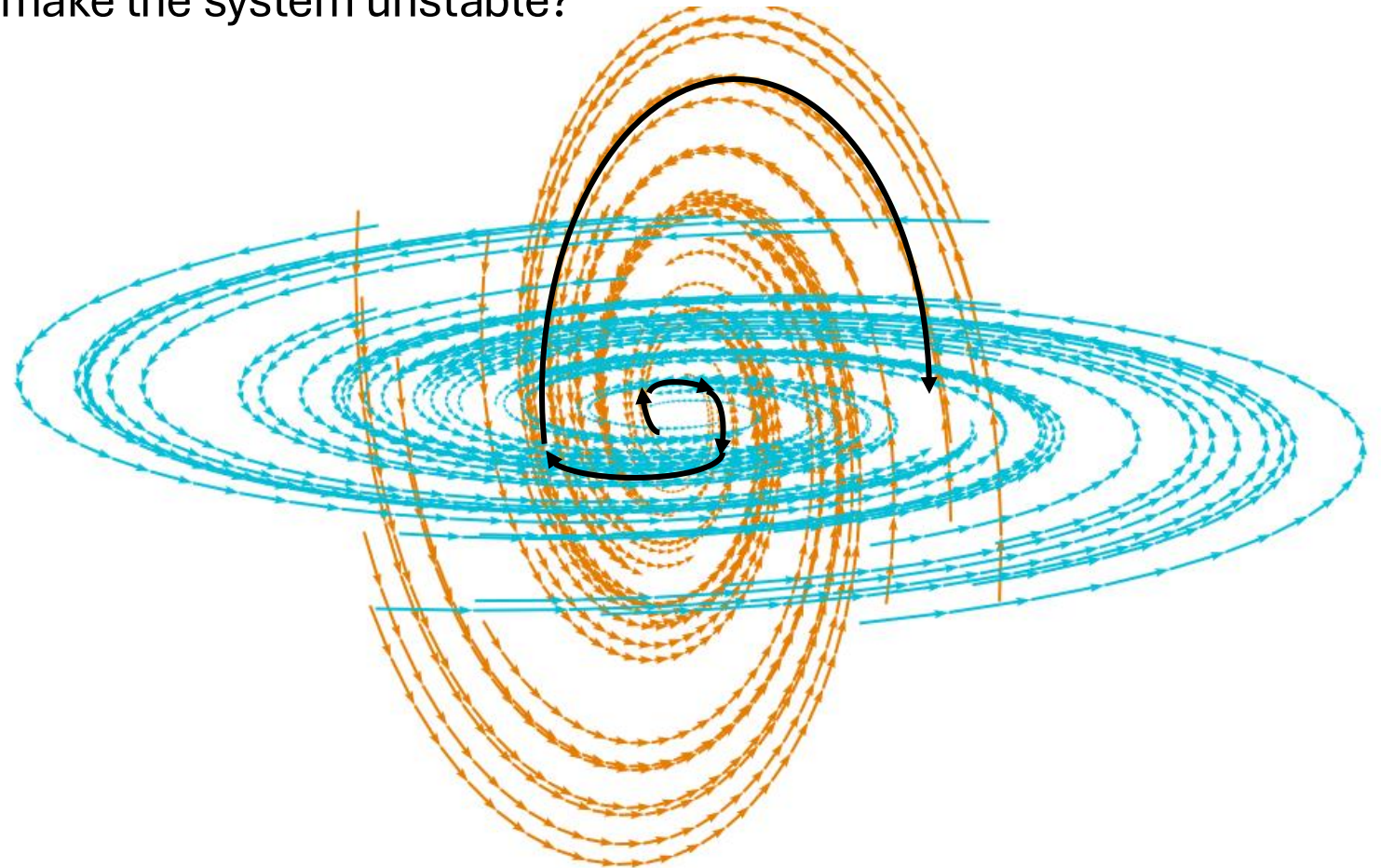
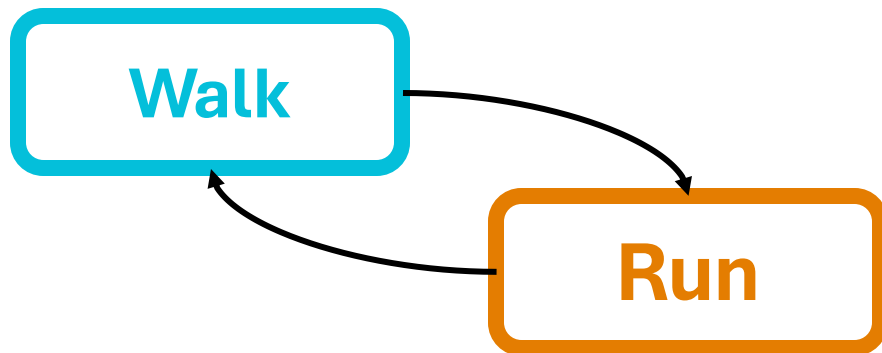
T is always ahead ...

Lesson: Mixing discrete transitions with continuous motion can be tricky!

Stability of a hybrid automaton

Each of the modes of a walking robot are asymptotically stable

Is it possible to switch between them to make the system unstable?



Special kinds of executions

- **Infinite:** Infinite sequence of transitions and trajectories

$\tau_0 a_1 \tau_1 a_2 \tau_2 \dots$

- **Closed:** Finite with final trajectory with closed domain

$\tau_0 a_1 \tau_1 a_2 \tau_2 \dots \tau_k$ and $\tau_k.dom = [0, T]$

- **Admissible:** Infinite duration

- May or may not be infinite

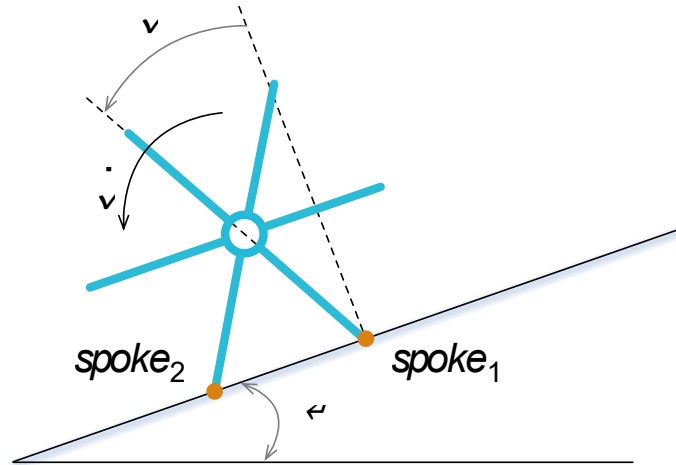
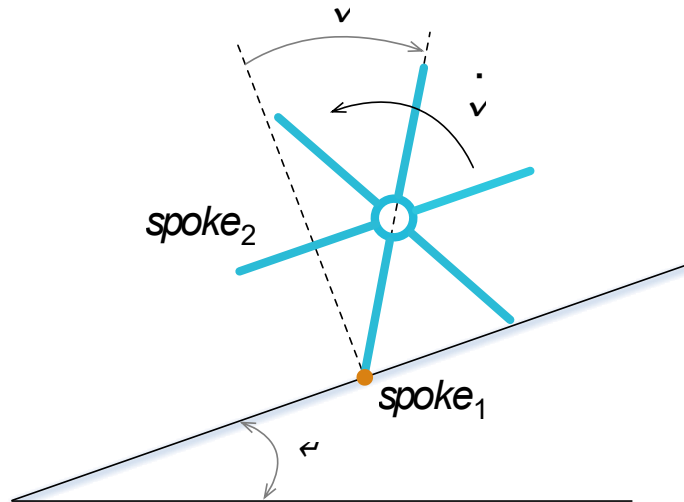
• $\tau_0 a_1 \tau_1 a_2 \tau_2 \dots$

• $\tau_0 a_1 \tau_1 a_2 \tau_2 \dots \tau_k$ with $\tau_k.dom = [0, \infty)$

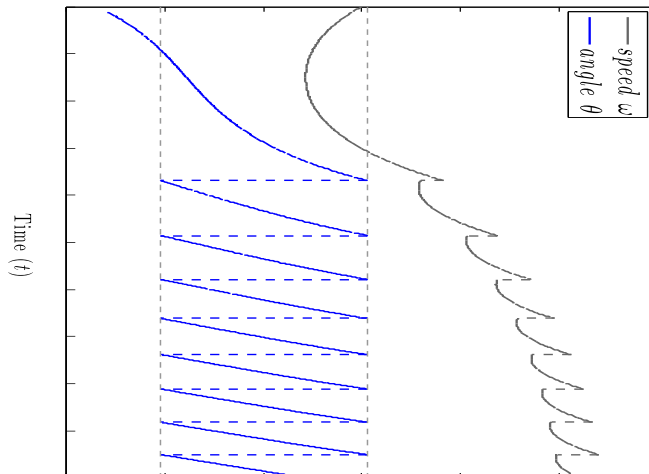
- **Zeno:** Infinite but not admissible

- Infinite number of transitions in finite time

Rimless wheel



θ, ω



automaton RimlessWheel(α, μ : Real, n : Nat)

const β : Real := $2 \pi / n$

type Spokes: enumeration [1,...,n]

actions

impact

variables

pivot: Spokes := 1

θ : Real := 0

ω : Real := 0

transitions

impact

pre $\theta \geq \beta/2$

eff pivot := pivot + 1 mod n

$\theta := \beta/2$

$\omega := \mu \omega$

trajectories

swing

evolve

$d(\theta) = \omega$

$d(\omega) = \sin(\theta + \alpha)$

invariant $\theta \leq \frac{\beta}{2}$

Invariants and reachability

A state x of $\mathcal{A}$ is **reachable** if $\exists$ an execution α with $\alpha.lstate = x$

$Reach_{\mathcal{A}}(\Theta)$ all reachable state from Θ

$Reach_{\mathcal{A}}(\Theta, T)$ states reachable within time T

$Reach_{\mathcal{A}}(\Theta, k)$ states reachable within k transitions

$Reach_{\mathcal{A}}(\Theta, T, k)$ states reachable up to time k transitions and time T

An invariant $I \subseteq val(X)$ is a set of states that contains $Reach_{\mathcal{A}}(\Theta)$

How to prove invariants of hybrid automata

Recall the theorem we used for proving I is an invariant for automaton $\mathcal{A} = \langle X, \Theta, A, D \rangle$

Theorem 7.1. Given an automaton $\mathcal{A} = \langle X, \Theta, A, \mathbf{D} \rangle$, if a set of states $I \subseteq \text{val}(X)$ satisfies the following:

- (Start condition) For any starting state $x \in \Theta, x \in I$ and
- (Transition closure) For any action $a \in A$, if $x \rightarrow_a x'$ and $x \in I$ then $x' \in I$, and

Then I is an inductive invariant of $\mathcal{A}$.

How to prove invariants of hybrid automata

Theorem 7.1. Given an HIOA $\mathcal{A} = \langle X, \Theta, A, \mathbf{D}, \mathbf{T} \rangle$, if a set of states $I \subseteq \text{val}(X)$ satisfies the following:

- (Start condition) For any starting state $x \in \Theta$, $x \in I$ and
- (Transition closure) For any action $a \in A$, if $x \rightarrow_a x'$ and $x \in I$ then $x' \in I$, and
- (Trajectory closure) For any trajectory $\tau \in \mathbf{T}$ if $\tau.fstate \in I$ then $\tau.lstate \in I$

Then I is an inductive invariant of $\mathcal{A}$.

How to prove invariants of hybrid automata

Theorem 7.1. Given an HIOA $\mathcal{A} = \langle X, \Theta, A, \mathbf{D}, \mathbf{T} \rangle$, if a set of states $I \subseteq \text{val}(X)$ satisfies the following:

- (Start condition) For any starting state $x \in \Theta$, $x \in I$ and
- (Transition closure) For any action $a \in A$, if $x \rightarrow_a x'$ and $x \in I$ then $x' \in I$, and
- (Trajectory closure) For any trajectory $\tau \in \mathbf{T}$ if $\tau.fstate \in I$ then $\tau.lstate \in I$

Then I is an inductive invariant of $\mathcal{A}$.

Proof. Consider any reachable state $x \in \text{Reach}_{\mathcal{A}}$. By the definition of a reachable state, there exists an execution α of $\mathcal{A}$ with $\alpha.lstate = x$. We proceed by induction on the length of the execution α . For the base case, α consists of a single starting state $x \in \Theta$, and, by the *start condition*, $x \in I$. For the inductive step, we consider two subcases:

Case 1: $\alpha = \alpha' a p(x)$, where $a \in A$ and $p(x)$ is a point trajectory at x .

By the induction hypothesis, we know that $\alpha'.lstate \in I$.

By invoking the *transition closure*, we obtain $x \in I$.

Case 2: $\alpha = \alpha' \tau$, where τ is a trajectory of $\mathcal{A}$ and $\tau.lstate = x$

By the *induction hypothesis*, $\alpha'.lstate \in I$ and by

invoking the *trajectory closure*, we deduce that $\tau.lstate = x \in I$

An application

automaton Bouncingball(c,h,g)

variables: x: Reals := h, v: Reals := 0

actions: bounce

transitions:

bounce

pre $x = 0 \wedge v < 0$

eff $v := -cv$

trajectories:

Loc1

evolve $d(x) = v; d(v) = -g$

invariant $x \geq 0$

Candidate invariant: “` stays above ground”

$$I_0: x \geq 0 \equiv \{u \in \text{val}(\{x, v\}) \mid u[x \geq 0]\}$$

Applying Theorem 7.1:

- Consider any initial state $u \in \Theta; u[x = h \geq 0]$
 - $u \in I_0$
- Consider any transition $u \rightarrow_{\text{bounce}} u'$
 - From precondition we know $u[x = 0]$; from effect we know $u'.x = u.x$ therefore $u'[x = 0 \geq 0]$
 - $u' \in I_0$
- Consider any trajectory $\tau \in T$
 - From mode invariant we know that for $\forall t \in \tau.\text{dom}, \tau(t)[x \geq 0]$
 - It follows that $\tau.\text{lstate}[x \geq 0]$
- What part of Bouncingball was used ? What could be changed?

An application

automaton Bouncingball(c,h,g)

variables: x: Reals := h, v: Reals := 0

actions: bounce

transitions:

bounce

pre $x = 0 \wedge v < 0$

eff $v := -cv$

trajectories:

Loc1

evolve $d(x) = v; d(v) = -g$

invariant $x \geq 0$

Candidate invariant: “` stays above ground and below h”

$$I_h: h \geq x \geq 0$$

Applying Theorem 7.1:

- Consider any initial state $\mathbf{u} \in \Theta$; $\mathbf{u}.x = h$
 - $\mathbf{u} \in I_h$
- Consider any transition $\mathbf{u} \rightarrow_{\text{bounce}} \mathbf{u}'$
 - From precondition we know $\mathbf{u}.x = 0$; from effect we know $\mathbf{u}'.x = \mathbf{u}.x$ therefore $\mathbf{u}'.x = 0$
 - $\mathbf{u}' \in I_h$
- Consider any trajectory $\tau \in T$
 - From mode invariant and inductive hypothesis we know that for $\forall t \in \tau.\text{dom}, \tau(t).x \geq 0$ **and** , $\tau(0).x \in [0, h]$ and that τ is a solution of $d(x) = v; d(v) = -g$
 - Is this adequate to infer $\tau.\text{lstate} \in I_h$?

Strengthened invariant

automaton Bouncingball(c,h,g)

variables: x: Reals := h, v: Reals := 0

k: Nat := 0

actions: bounce

transitions:

bounce

pre $x = 0 \wedge v < 0$

eff $v := -cv$; $k := k + 1$

trajectories:

Loc1

evolve $d(x) = v$; $d(v) = -g$

invariant $x \geq 0$

Candidate invariant: “` stays above ground and below h”

$$I_v: v^2 - 2g(hc^{2k} - x) = 0$$

Applying Theorem 7.1:

- Consider any initial state $u \in \Theta$; $u.x = h$; $u.k = 0$
 - $u \in I_v$
- **Exercise:** Finish the rest

Summary

- Theorem 7.1 gives a sufficient condition for proving **inductive** invariants
- Not all invariants are inductive
- We often have to **strengthen** invariants to make them inductive
- Read examples in Chapter 7

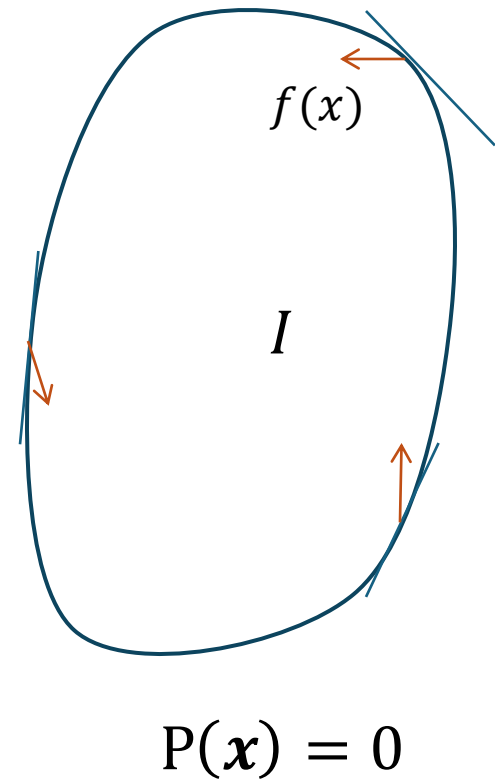
Sub-tangential conditions. Checking trajectory conditions without solving ODEs [Bhatia and Szegö 67]

How to check trajectory closure: for any $\tau \in T$ if $\tau.fstate \in I$ then $\tau.lstate \in I$?

Use a Lyapunov function for each mode. How? Exercise.

Lemma. Consider the ODE $\dot{x} = f(x)$ for state variable x , describing T . Let I be a compact set containing the initial set Θ . Boundary of I is defined by $P(x) = 0$. Then, I is an inductive invariant of the above ODE if for each x , on the boundary $P(x) = 0$, $f(x)$ is pointing inwards from the boundary. That is

$$\frac{\partial P(x)}{\partial x} \cdot f(x) \geq 0.$$



Related to the notions of Barrier Certificates and more general Control Barrier Functions.

Proving invariance using subtangential condition

Lemma. Consider $\dot{x} = f(x)$ and let $P: \mathbb{R}^n \rightarrow \mathbb{R}$ be a continuous function such that $I = \{x | P(x) \leq 0\}$ is compact and contains. $\partial I := \{x | P(x) = 0\}$ is the boundary of I and P is differentiable over ∂I . If

Sub-tangential condition $\frac{\partial P(x)}{\partial x} \cdot f(x) \leq 0 \quad \forall x \in \partial I$.

Then I is an invariant

Proof. Fix $x(0) \in \Theta \subseteq I$. We must show $\forall t \geq 0, x(t) \in I$

Define $\psi(t) = P(x(t))$

$\psi(0) = P(x(0)) \leq 0 \quad [x(0) \in \Theta \subseteq I, P(x(0)) \leq 0]$

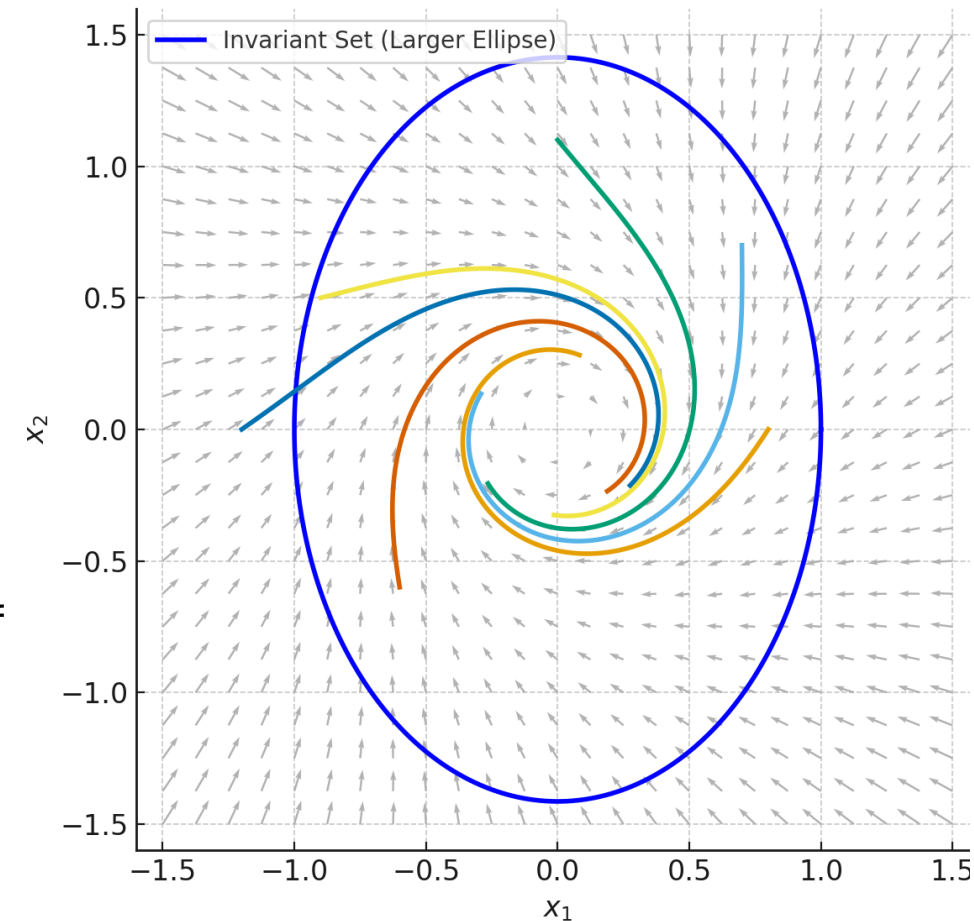
SFTC trajectory from $x(0)$ leaves I at some point t_1

That is, $\psi(t_1) = P(x(t_1)) > 0$

By continuity of ψ , there exists earliest $t^* \in (0, t_1)$ such that $\psi(t) \leq 0$ f $t < t^*$, $\psi(t^*) = 0 \Rightarrow x(t^*) \in \partial I$ and $\psi(t) > 0$ for some $t > t^*$

$$\dot{\psi}(t^*) = \frac{d}{dt} P(x(t)) \Big|_{t=t^*} = \frac{\partial P(x(t^*))}{\partial x} \cdot f(x(t^*)) \leq 0$$

$\psi(t^*)$ is nonincreasing and it is not possible for $\psi(t) > 0$ for $t > t^*$.



Lemma. Consider $\dot{x} = f(x)$ and let $P: \mathbb{R}^n \rightarrow \mathbb{R}$ be a continuous function such that $I = \{x | P(x) \leq 0\}$ is compact and contains. $\partial I := \{x | P(x) = 0\}$ is the boundary of I and P is differentiable over ∂I . If

Sub-tangential condition $\frac{\partial P(x)}{\partial x} \cdot f(x) \leq 0 \quad \forall x \in \partial I$.

Then I is an invariant

Example.

$$\dot{x}_1 = x_2 - x_1(x_1^2 + x_2^2) \quad \dot{x}_2 = -x_1 - x_2(x_1^2 + x_2^2)$$

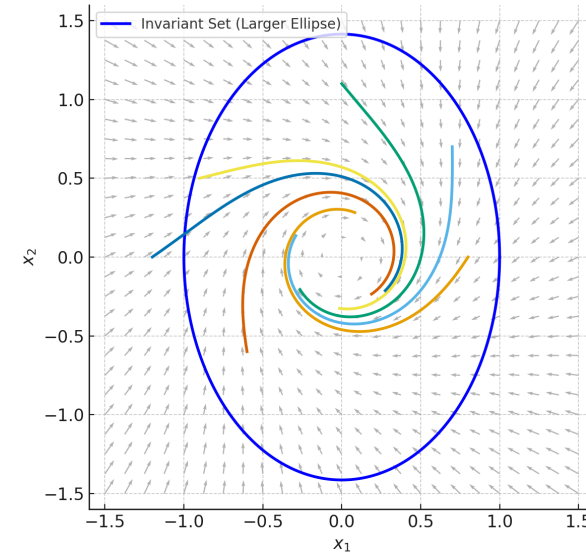
$$I = \{x \in \mathbb{R}^2 | P(x) \leq 0\} \text{ where } P(x) = 2x_1^2 + x_2^2 - 1$$

$$\nabla P = \frac{\partial P}{\partial x} = \begin{bmatrix} 4x_1 \\ 2x_2 \end{bmatrix}$$

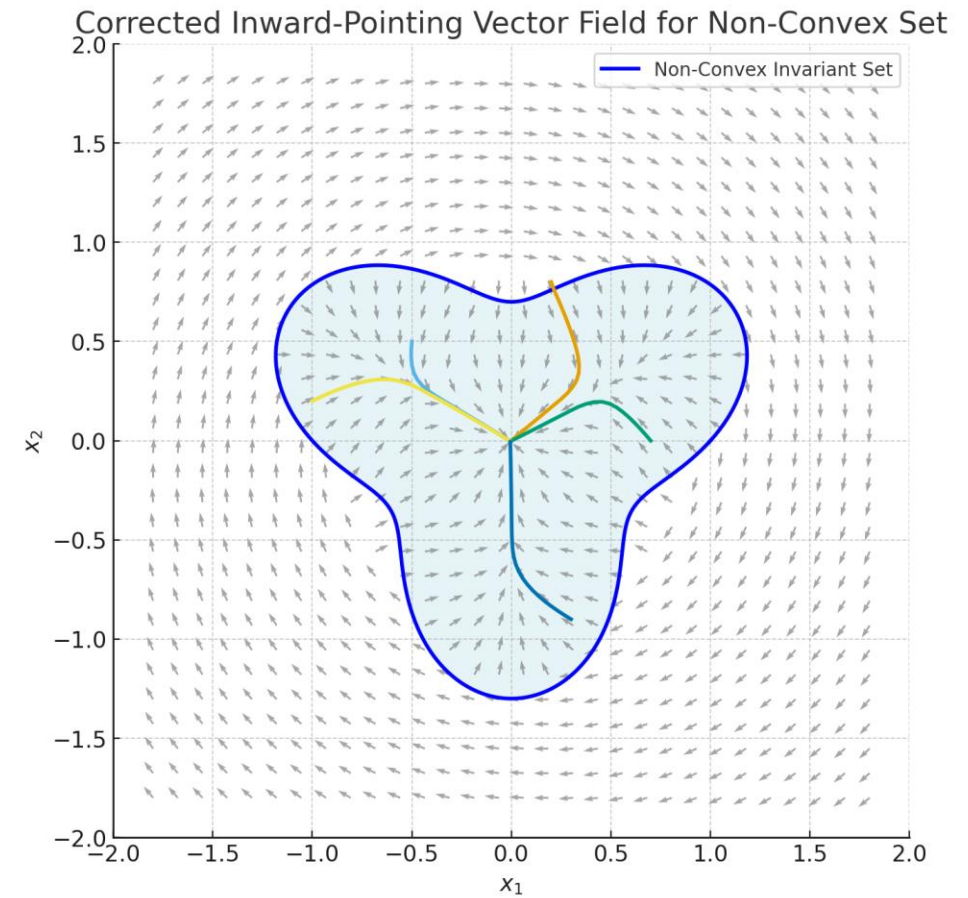
$$\begin{aligned} \nabla P \cdot f(x) &= \begin{bmatrix} 4x_1 \\ 2x_2 \end{bmatrix} \cdot [x_2 - x_1(x_1^2 + x_2^2) \quad -x_1 - x_2(x_1^2 + x_2^2)] \\ &= 4x_1(x_2 - x_1(x_1^2 + x_2^2)) + 2x_2(-x_1 - x_2(x_1^2 + x_2^2)) \\ &= 4x_1x_2 - 4x_1^2r^2 - 2x_2x_1 - 2x_2^2r^2 \\ &= 2x_1x_2 - 2r^2(2x_1^2 + x_2^2) \end{aligned}$$

$$\text{On the boundary } 2x_1^2 + x_2^2 = 1, \nabla P \cdot f(x) = 2x_1x_2 - 2r^2 = 2x_1x_2 - 2(x_1^2 + x_2^2)$$

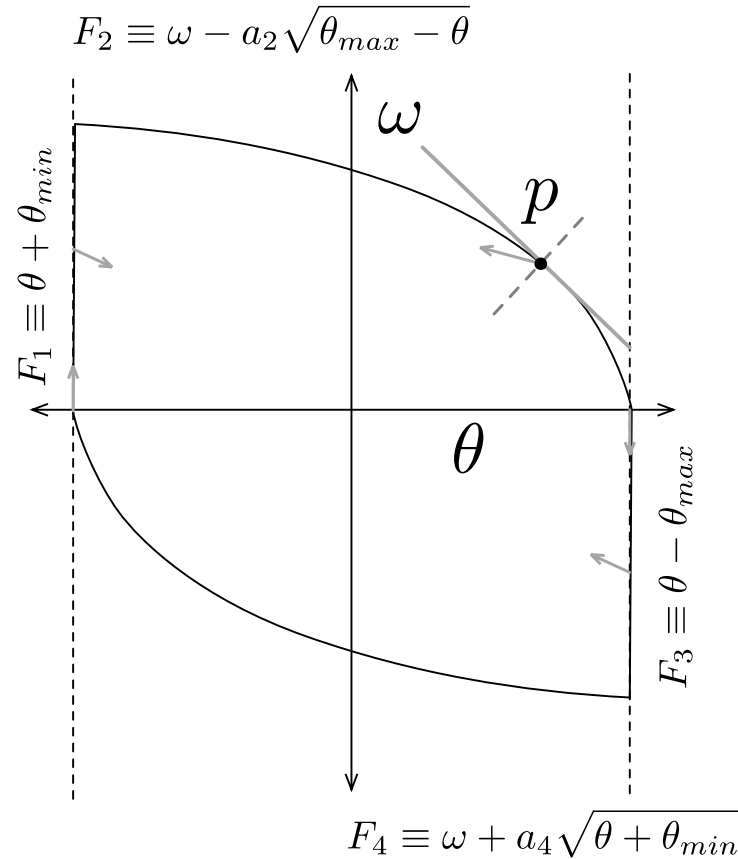
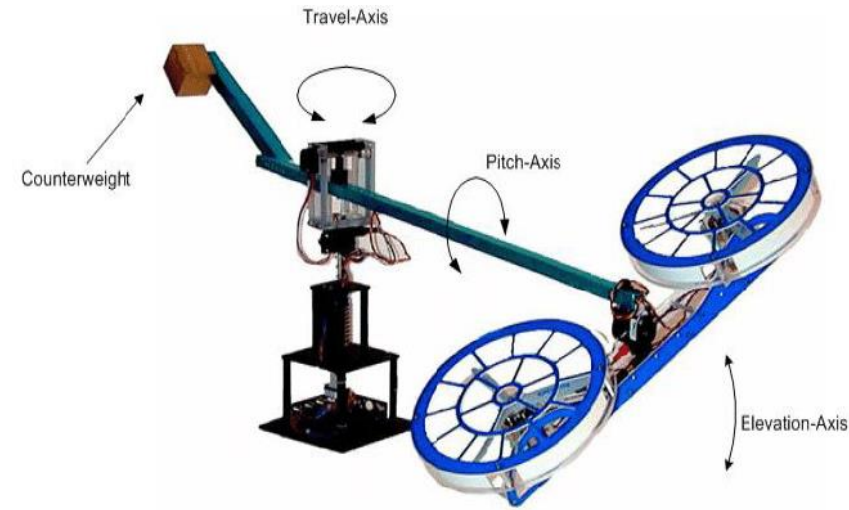
$$\text{Completing the square: } \nabla P \cdot f(x) = 2x_1x_2 - 2(x_1^2 + x_2^2) = -2 \left(\frac{3}{4}x_1^2 + \left(x_2 - \frac{1}{2}x_1\right)^2 \right) \leq 0$$



Another example



Piece-wise sub-tangential conditions



If the boundary is defined in pieces, for example, one for each mode, then check the condition for each piece separately

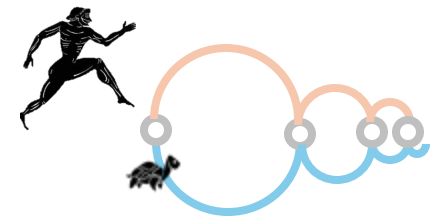
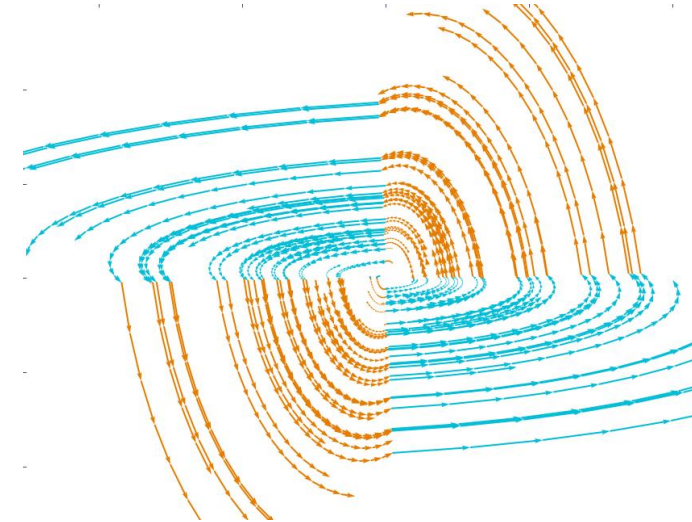
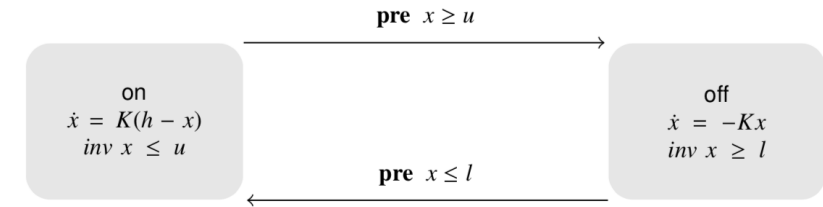
Summary

Interesting features in hybrid models

- Interaction between transitions, trajectories, and mode invariants can lead to extra sources of nondeterminism; can be removed with urgency
- Mode switches can make the system unstable
- Zeno behavior

Inductive invariance proofs work with additional condition for trajectory closure

- Trajectory closure can be checked with subtangential condition, Lyapunov functions, Barrier certificates
- This requires first finding an invariant/barrier; very active research on learning neural barriers by minimizing a loss representing violation of gradient conditions



Special Classes of Hybrid Automata

Algorithmic analysis of (Alur-Dill's) Timed Automata[1]

Timed Automata are a restricted class of hybrid automata with only clock variables which are amenable to automatic verification

[1] Rajeev Alur and David L. Dill. [A theory of timed automata](#). Theoretical Computer Science, 126:183-235, 1994.

Clocks and Clock Constraints

- A **clock variable** x is a continuous (analog) variable of type real such that along any trajectory τ of x , for all $t \in \tau.\text{dom}$, $(\tau \downarrow x)(t) = t$.
- For a set X of clock variables, the set $\Phi(X)$ of **integral clock constraints** are expressions defined by the syntax:

$$g ::= x \leq q \mid x \geq q \mid \neg g \mid g_1 \wedge g_2$$

where $x \in X$ and $q \in \mathbb{Z}$

- Examples: $x = 10$; $x \in [2, 5)$; true are valid clock constraints
- What do clock constraints look like?
- Semantics of clock constraints $[g]$

Integral Timed Automata

Definition. An **integral timed automaton** is a HIOA $\mathcal{A} = \langle V, \Theta, A, \mathcal{D}, \mathcal{T} \rangle$ where

- $V = X \cup \{l\}$, where X is a set of n clocks and l is a discrete state variable of finite type L
- A is a finite set
- $\mathcal{D}$ is a set of transitions such that
 - The guards are described by clock constraints $\Phi(X)$
 - $\langle x, l \rangle - a \rightarrow \langle x', l' \rangle$ implies either $x' = x$ or $x = 0$
- $\mathcal{T}$ set of clock trajectories for the clock variables in X

Example: Light switch

Math Formulation

automaton Switch

variables

internal $x, y: \text{Real} := 0, \text{loc}: \{\text{on}, \text{off}\} := \text{off}$

transitions

internal push

pre $x \geq 2$

eff if $\text{loc} = \text{on}$ **then** $x := 0$

else $x, y := 0; \text{loc} := \text{off}$

internal pop

pre $y = 15 \wedge \text{loc} = \text{off}$

eff $x := 0$

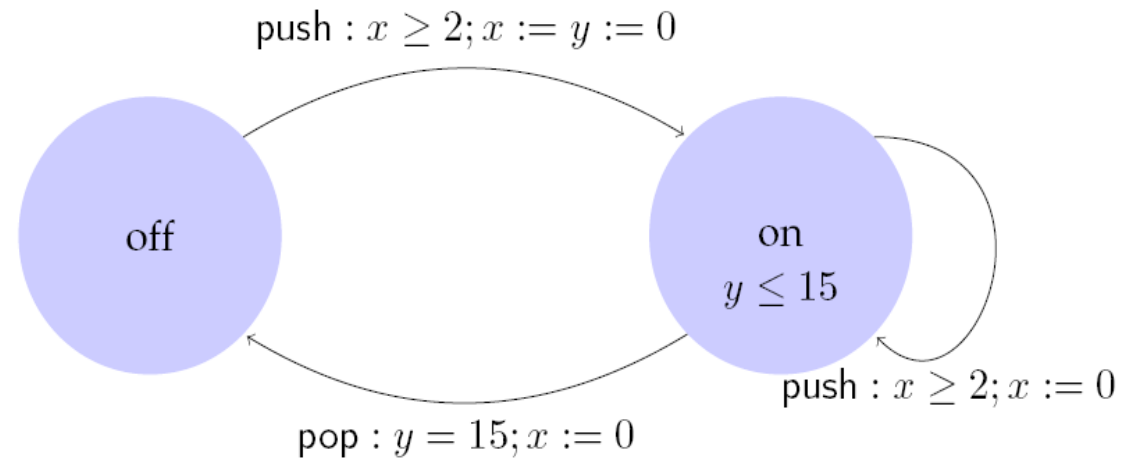
trajectories

invariant $\text{loc} = \text{off} \Rightarrow y \leq 15$

evolve $d(x) = 1; d(y) = 1$

Description

Switch can be turned on whenever at least 2 time units have elapsed since the last turn on. Switches off automatically 15 time units after the last on.



Control State (Location) Reachability Problem

- Given an ITA $\mathcal{A}$, check if a particular (discrete) control state is reachable from the initial states
- Why is control state reachability (CSR) good enough?
- This problem is decidable [Alur Dill]
- Key idea:
 - Construct a finite automaton that is **a time-abstract bisimilar** to the ITA (behaves identically with respect to control state reachability)
 - Check reachability of FSM

An equivalence relation with a finite quotient

- Under what conditions do two states x_1 and x_2 of the automaton $\mathcal{A}$ behave identically with respect to control state reachability (CSR)?
 - When do they satisfy the same set of clock constraints?
 - When would they continue to satisfy the same set of clock constraints?
- $x_1.\text{loc} = x_2.\text{loc}$ and
- x_1 and x_2 satisfy the same set of clock constraints
 - For each clock y $\text{int}(x_1.y) = \text{int}(x_2.y)$ or $\text{int}(x_1.y) \geq c_{\mathcal{A}y}$ and $\text{int}(x_2.y) \geq c_{\mathcal{A}y}$. ($c_{\mathcal{A}y}$ is the maximum clock guard of y)
 - For each clock y with $x_1.y \leq c_{\mathcal{A}y}$, $\text{frac}(x_1.y) = 0$ iff $\text{frac}(x_2.y) = 0$
 - For any two clocks y and z with $x_1.y \leq c_{\mathcal{A}y}$ and $x_1.z \leq c_{\mathcal{A}z}$, $\text{frac}(x_1.y) \leq \text{frac}(x_1.z)$ iff $\text{frac}(x_2.y) \leq \text{frac}(x_2.z)$
- **Lemma.** This is an equivalence relation on $\text{val}(V)$ the states of $\mathcal{A}$
- The partition of $\text{val}(V)$ induced by this relation is called clock regions

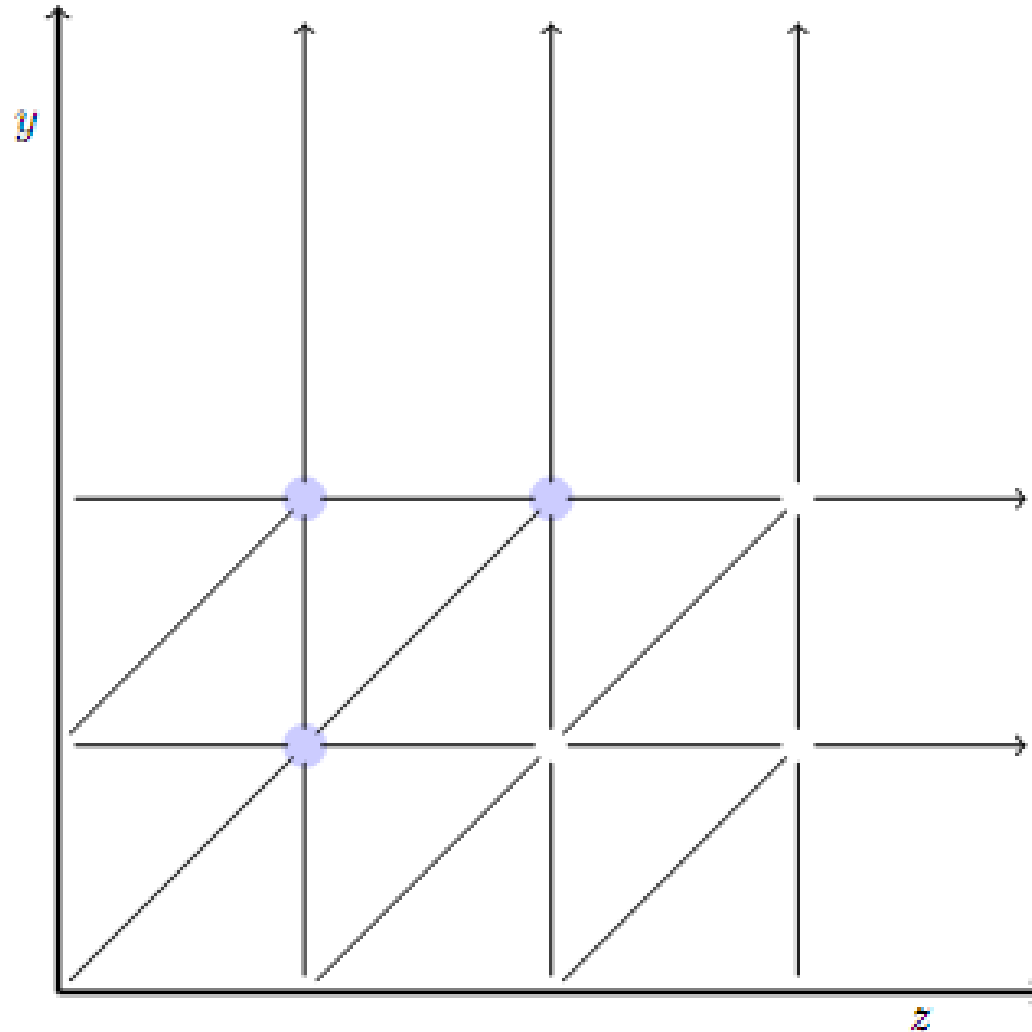
What do the clock regions look like?

Example of Two Clocks

$$X = \{y, z\}$$

$$c_{\mathcal{A}y} = 2$$

$$c_{\mathcal{A}z} = 3$$



Complexity

Lemma. The number of clock regions is bounded by $|X|! 2^{|X|} \prod_{z \in X} (2c_{\mathcal{A}_z} + 2)$.

Region automaton $R(\mathcal{A})$

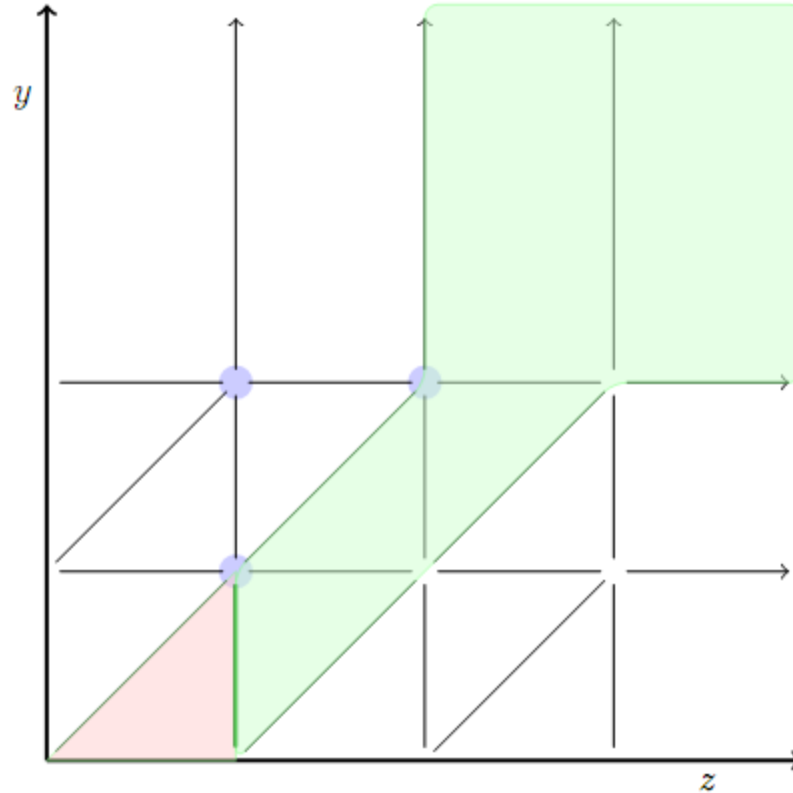
Given an ITA $\mathcal{A} = \langle V, \Theta, \mathcal{D}, \mathcal{T} \rangle$, we construct the corresponding **Region Automaton** $R(\mathcal{A}) = \langle Q_R, \Theta_R, D_R \rangle$ such that (i) $R(\mathcal{A})$ visits the same set of locations (but does not have timing information) and (ii) $R(\mathcal{A})$ is finite state machine.

- ITA (clock constants) defines a set of clock regions, say $C_{\mathcal{A}}$. The set of states $Q_R = C_{\mathcal{A}} \times L$
- $Q_0 \subseteq Q$ is the set of states contain initial set Θ of $\mathcal{A}$
- D : We add the transitions between Q (regions)
 - **Time successors**: Consider two clock regions γ and γ' , we say that γ' is a time successor of γ if there exists a trajectory of ITA starting from γ that ends in γ'
 - **Discrete transitions**: Same as the ITA

Theorem. A location of ITA $\mathcal{A}$ is reachable iff it is also reachable in $R(\mathcal{A})$.

(we say that $R(\mathcal{A})$ is *time abstract bisimilar* to $\mathcal{A}$)

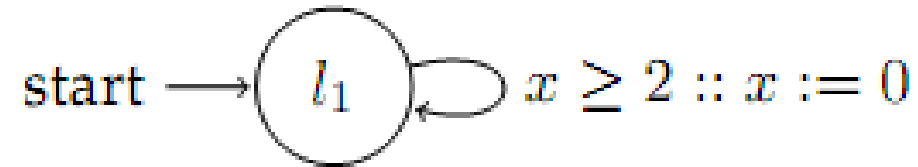
Time successors



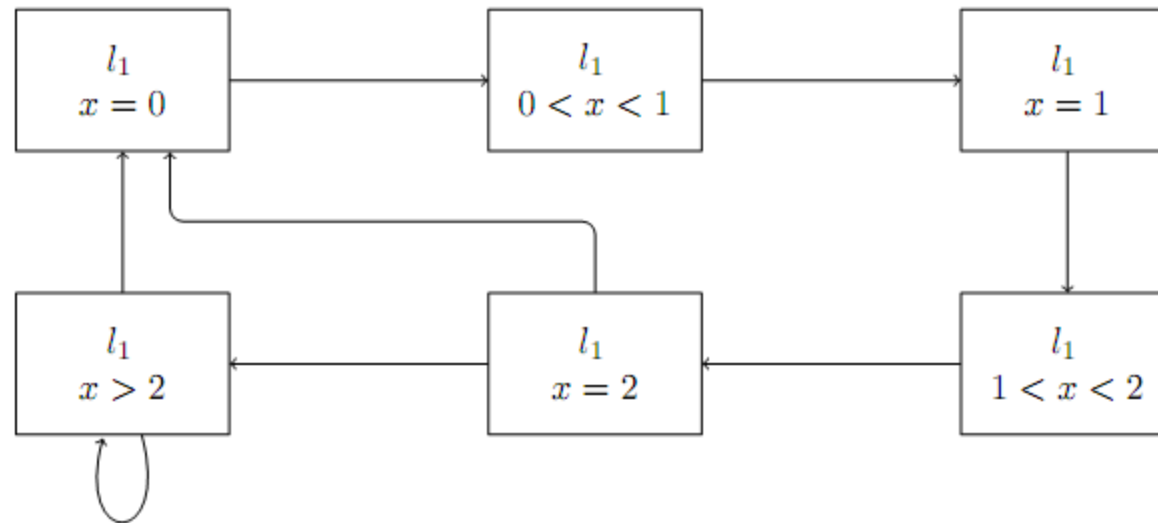
The clock regions in blue are time successors of the clock region in red.

Example 1: Region Automata

ITA

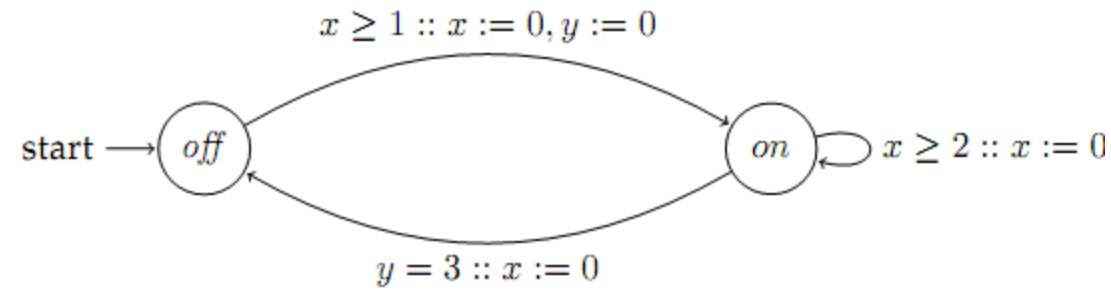


Corresponding FA

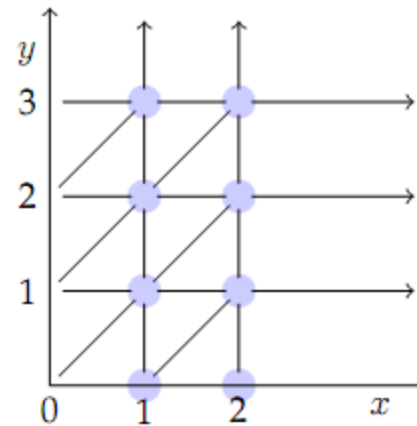


Example 2

ITA



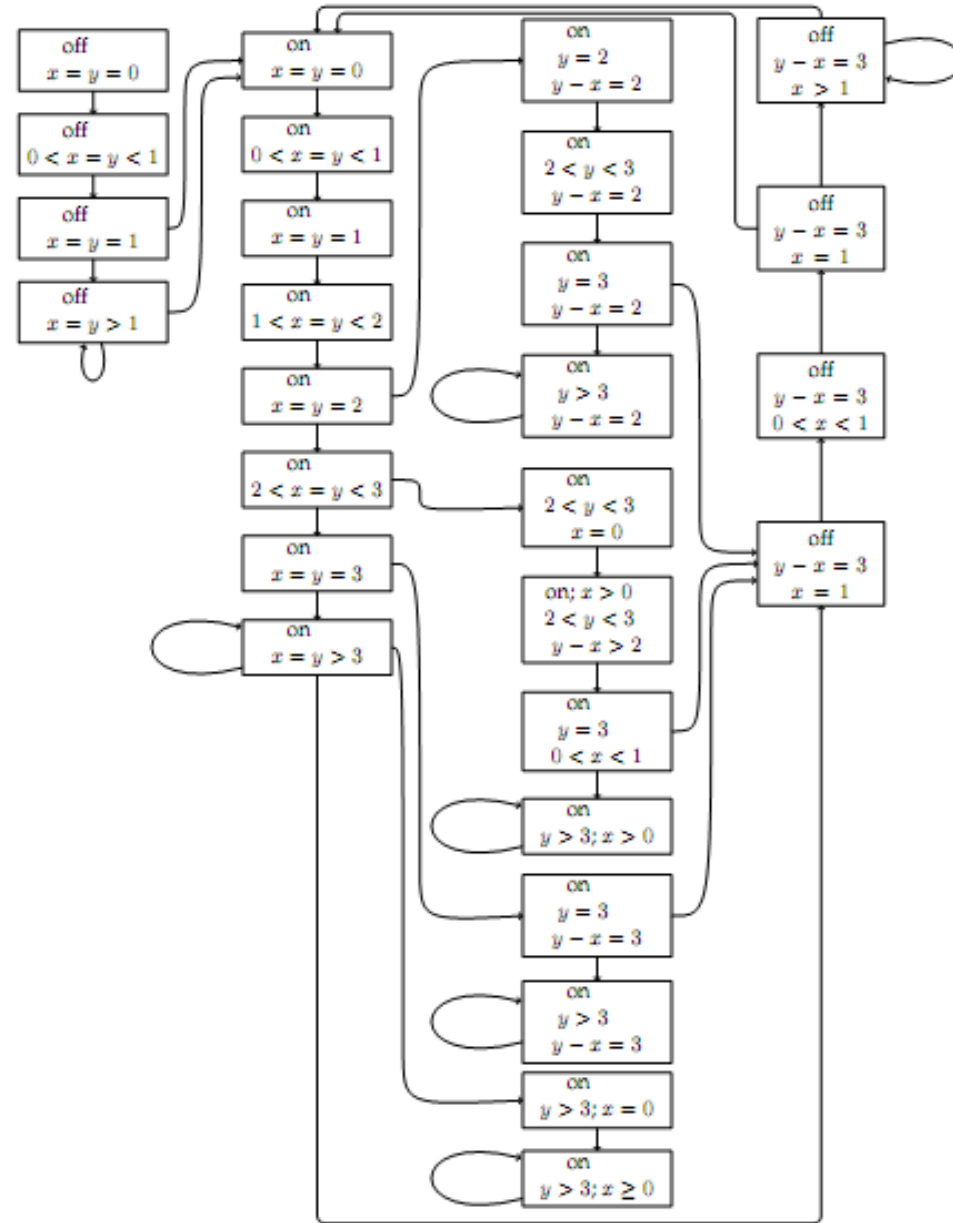
Clock
Regions



Corresponding FA

$$|X|! 2^{|X|} \prod_{z \in X} (2c_{\mathcal{A}z} + 2)$$

Drastically increasing with
the number of clocks



Special Classes of Hybrid Automata

- Timed Automata ←
- Rational time automata
- Multirate automata
- Rectangular Initialized HA
- Rectangular HA
- Linear HA
- Nonlinear HA

Clocks and **Rational** Clock Constraints

- A **clock variable** x is a continuous (analog) variable of type real such that along any trajectory τ of x , for all $t \in \tau.\text{dom}$, $(\tau \downarrow x)(t) = t$.
- For a set X of clock variables, the set $\Phi(X)$ of **rational clock constraints** are expressions defined by the syntax:
$$g ::= x \leq q \mid x \geq q \mid \neg g \mid g_1 \wedge g_2$$

where $x \in X$ and $q \in \mathbb{Q}$
- Examples: $x = 10.125$; $x \in [2.99, 5)$; true are valid rational clock constraints
- Semantics of clock constraints $[g]$

Step 1. Rational Timed Automata

- **Definition.** A *rational timed automaton* is a HA $\mathcal{A} = \langle V, \Theta, A, \mathcal{D}, \mathcal{T} \rangle$ where
 - $V = X \cup \{loc\}$, where X is a set of n clocks and l is a discrete state variable of finite type $\mathbb{L}$
 - A is a finite set
 - $\mathcal{D}$ is a set of transitions such that
 - The guards are described by **rational** clock constraints $\Phi(X)$
 - $\langle x, l \rangle - a \rightarrow \langle x', l' \rangle$ implies either $x' = x$ or $x = 0$
 - $\mathcal{T}$ set of clock trajectories for the clock variables in X

Example: Rational Light switch

Switch can be turned on whenever at least 2.25 time units have elapsed since the last turn off or on. Switches off automatically 15.5 time units after the last on.

automaton Switch

internal push; pop

variables

internal $x, y: \text{Real} := 0$, $\text{loc}:\{\text{on}, \text{off}\} := \text{off}$

transitions

push

pre $x \geq 2.25$

eff if $\text{loc} = \text{on}$ **then** $y := 0$ **fi**; $x := 0$; $\text{loc} := \text{off}$

pop

pre $y = 15.5 \wedge \text{loc} = \text{off}$

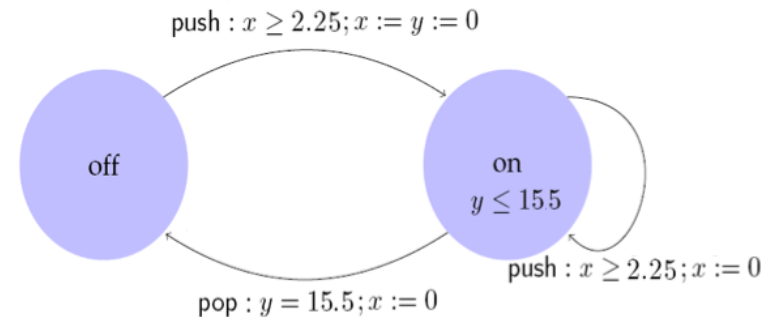
eff $x := 0$

trajectories

invariant $\text{loc} = \text{on} \vee \text{loc} = \text{off}$

stop when $y = 15.5 \wedge \text{loc} = \text{off}$

evolve $d(x) = 1$; $d(y) = 1$



Control State (Location) Reachability Problem

- Given an RTA, check if a particular location is reachable from the initial states
- Is problem decidable?
- Yes
- Key idea:
 - Construct a ITA that is time-abstract bisimilar to the given RTA
 - Check CSR for ITA

Construction of ITA from RTA

- Multiply all rational constants by a factor q that make them integral
- Make $d(x) = q$ for all the clocks
- RTA Switch is bisimilar to ITA lswitch
- Simulation relation R is given by
- $(\mathbf{u}, \mathbf{s}) \in R$ iff $\mathbf{u}.x = 4 \mathbf{s}.x$ and $\mathbf{u}.y = 4 \mathbf{s}.y$

automaton lSwitch

internal push; pop

variables

internal $x, y: \text{Real} := 0, \text{loc}: \{\text{on}, \text{off}\} := \text{off}$

transitions

push

pre $x \geq 9$

eff if $\text{loc} = \text{on}$ **then** $y := 0$ **fi**; $x := 0; \text{loc} := \text{off}$

pop

pre $y = 62 \wedge \text{loc} = \text{off}$

eff $x := 0$

trajectories

invariant $\text{loc} = \text{on} \vee \text{loc} = \text{off}$

stop when $y = 62 \wedge \text{loc} = \text{off}$

evolve $d(x) = 4; d(y) = 4$

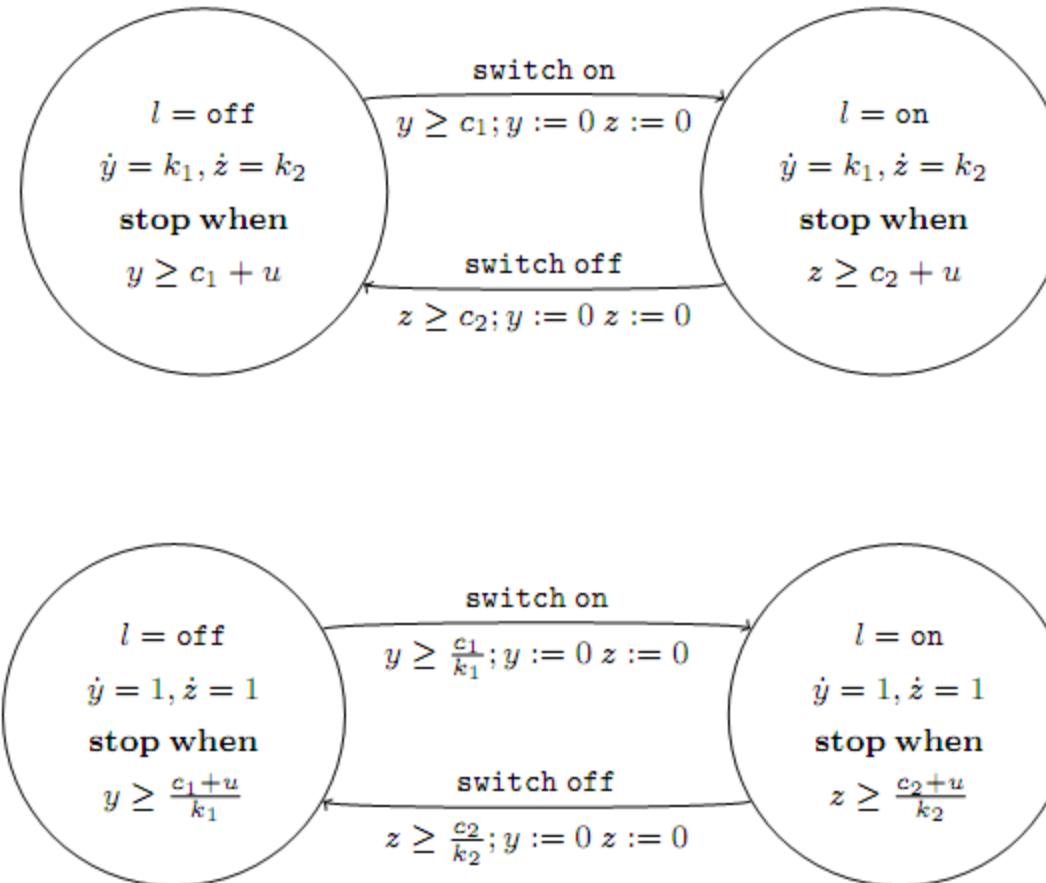
Step 2. Multi-Rate Automaton

- **Definition.** A **multirate automaton** is $\mathcal{A} = \langle V, Q, \Theta, A, \mathcal{D}, \mathcal{T} \rangle$ where
 - $V = X \cup \{loc\}$, where X is a set of n **continuous variables** and loc is a discrete state variable of finite type $\mathbb{L}$
 - A is a finite set of actions
 - $\mathcal{D}$ is a set of transitions such that
 - The guards are described by **rational** clock constraints $\Phi(X)$
 - $\langle x, l \rangle - a \rightarrow \langle x', l' \rangle$ implies either $x' = c$ or $x' = x$
 - $\mathcal{T}$ set of trajectories such that
 - for each variable $x \in X \exists k$ such that $\tau \in \mathcal{T}, t \in \tau.dom$
$$\tau(t).x = \tau(0).x + k t$$

Control State (Location) Reachability Problem

- Given an MRA, check if a particular location is reachable from the initial states
- Is problem is decidable? Yes
- Key idea:
 - Construct a RTA that is bisimilar to the given MRA

Example: Multi-rate to rational TA



Step 3. Rectangular HA

Definition. An **rectangular hybrid automaton (RHA)** is a HA $\mathcal{A} = \langle V, A, \mathcal{T}, \mathcal{D} \rangle$ where

- $V = X \cup \{loc\}$, where X is a set of n **continuous variables** and loc is a discrete state variable of finite type $\mathbb{L}$
- A is a finite set
- $\mathcal{T} = \bigcup_{\ell} \mathcal{T}_{\ell}$ set of trajectories for X
 - For each $\tau \in \mathcal{T}_{\ell}, x \in X$ either (i) $d(x) = k_{\ell}$ or (ii) $d(x) \in [k_{\ell_1}, k_{\ell_2}]$
 - Equivalently, (i) $\tau(t)[x] = \tau(0)[x] + k_{\ell}t$
(ii) $\tau(0)[x] + k_{\ell_1}t \leq \tau(t)[x] \leq \tau(0)[x] + k_{\ell_2}t$
- $\mathcal{D}$ is a set of transitions such that
 - Guards are described by **rational** clock constraints
 - $\langle x, l \rangle \rightarrow_a \langle x', l' \rangle$ implies $x' = x$ or $x' \in [c_1, c_2]$

CSR Decidable for RHA?

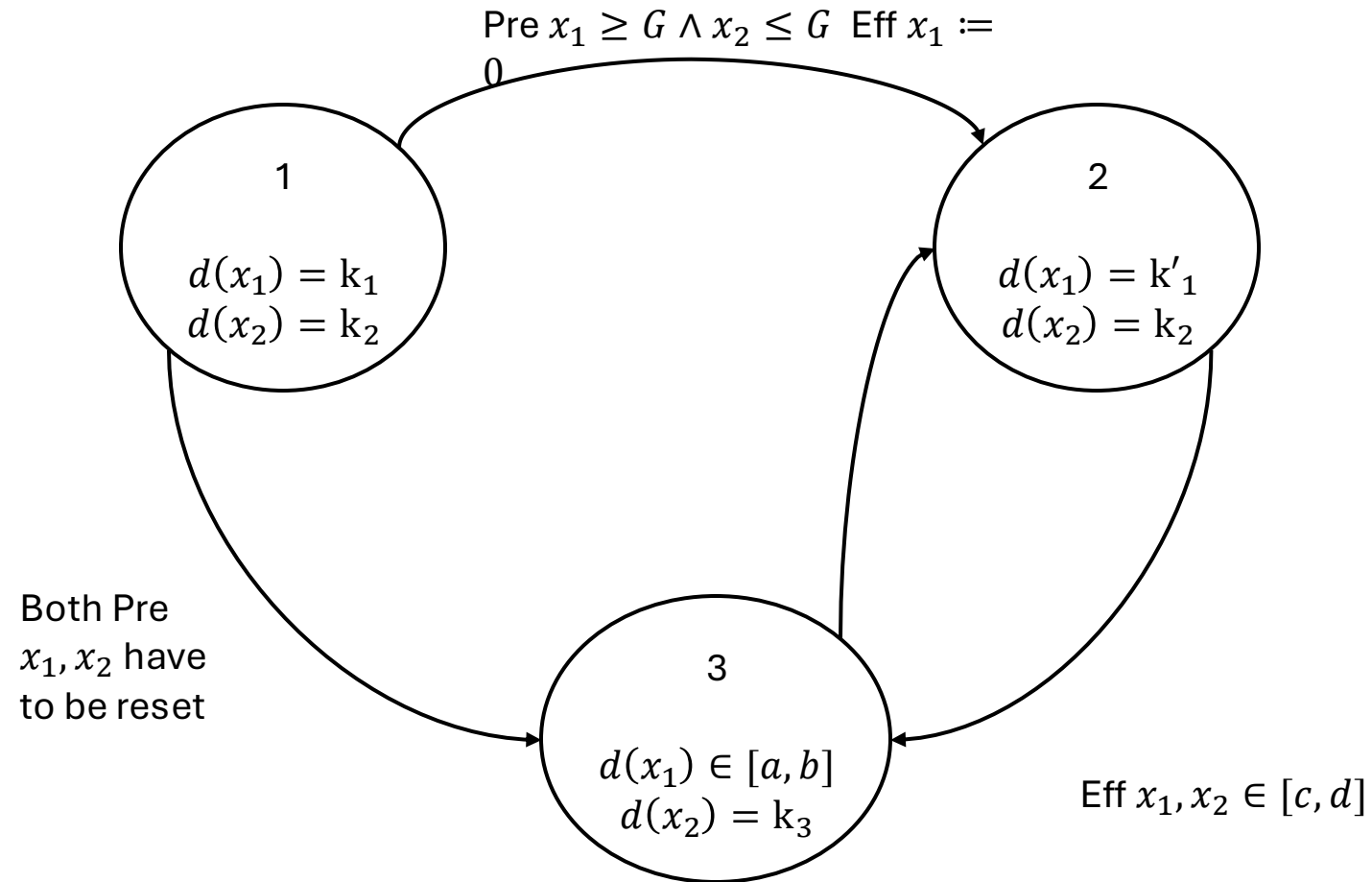
- Given an RHA, check if a particular location is reachable from the initial states?
- Is this problem decidable? **No**
 - **[Henz95]** Thomas Henzinger, Peter Kopke, Anuj Puri, and Pravin Varaiya. [What's Decidable About Hybrid Automata?. Journal of Computer and System Sciences, pages 373–382. ACM Press, 1995.](#)
 - CSR for RHA reduction to Halting problem for 2 counter machines
 - Halting problem for 2CM known to be undecidable
 - Reduction in next lecture

Step 4. Initialized Rectangular HA

Definition. An *initialized rectangular hybrid automaton (IRHA)* is a RHA $\mathcal{A}$ where

- $V = X \cup \{loc\}$, where X is a set of n continuous variables and $\{loc\}$ is a discrete state variable of finite type $\mathbb{L}$
- A is a finite set
- $\mathcal{T} = \bigcup_{\ell} \mathcal{T}_{\ell}$ set of trajectories for X
 - For each $\tau \in \mathcal{T}_{\ell}, x \in X$ either (i) $d(x) = k_{\ell}$ or (ii) $d(x) \in [k_{\ell_1}, k_{\ell_2}]$
 - Equivalently, (i) $\tau(t)[x] = \tau(0)[x] + k_{\ell}t$
(ii) $\tau(0)[x] + k_{\ell_1}t \leq \tau(t)[x] \leq \tau(0)[x] + k_{\ell_2}t$
- $\mathcal{D}$ is a set of transitions such that
 - Guards are described by **rational** clock constraints
 - $\langle x, l \rangle \rightarrow_a \langle x', l' \rangle$ implies if dynamics changes from ℓ to ℓ' then $x' \in [c_1, c_2]$, otherwise $x' = x$

Example: Rectangular Initialized HA



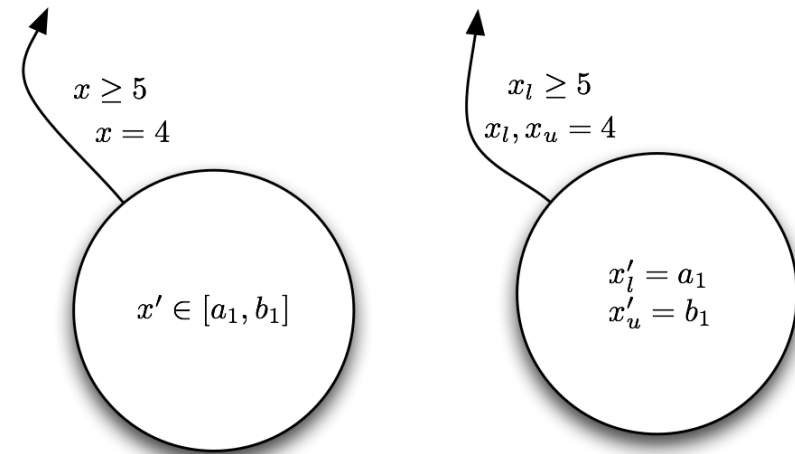
CSR Decidable for IRHA?

- Given an IRHA, check if a particular location is reachable from the initial states
- Is this problem decidable? **Yes**
- Key idea:
 - Construct a $2n$ -dimensional **initialized multi-rate** automaton that is bisimilar to the given IRHA
 - Construct a ITA that is bisimilar to the Singular TA

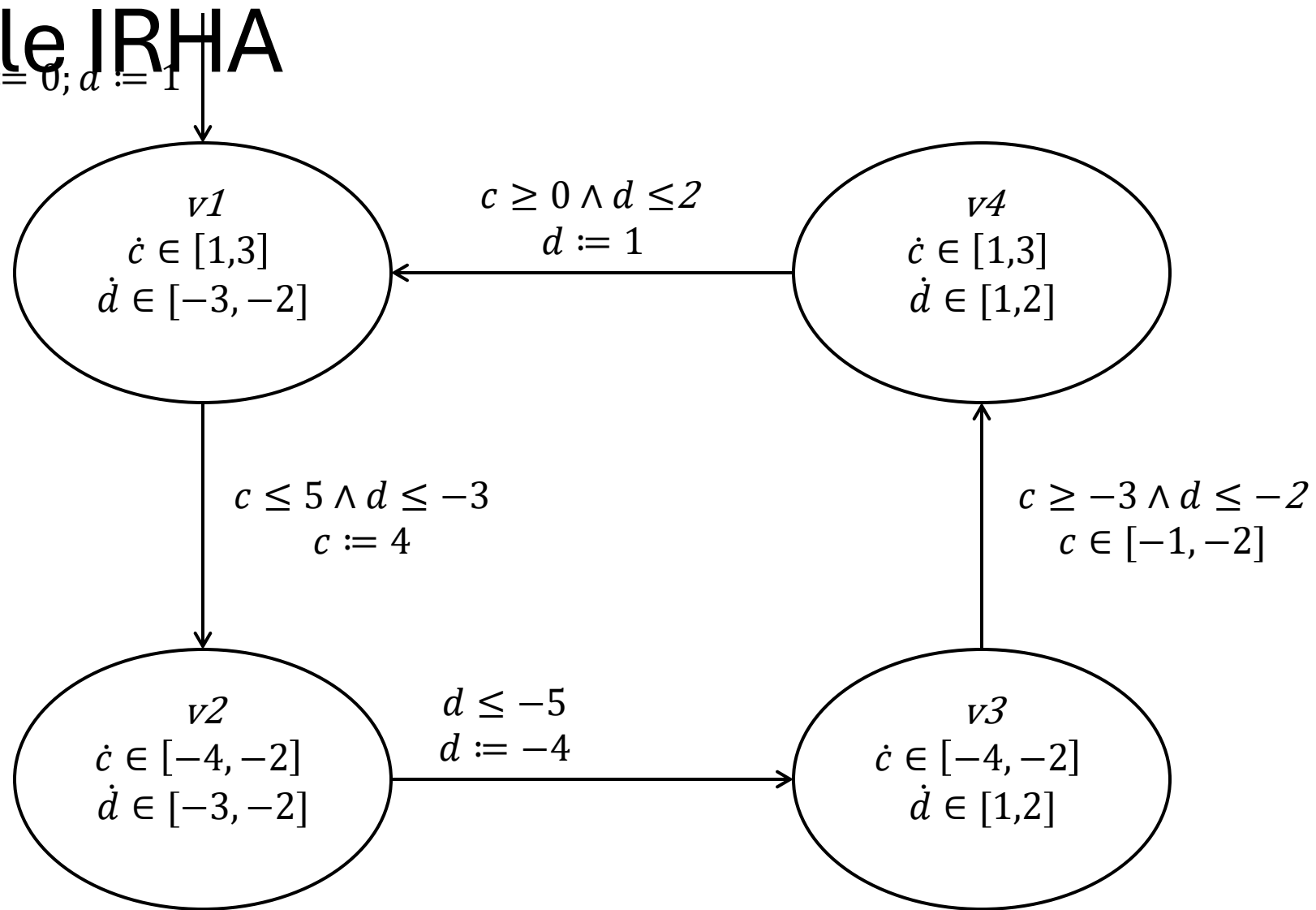
From IRHA to Singular HA conversion

For every variable create two variables---tracking the upper and lower bounds

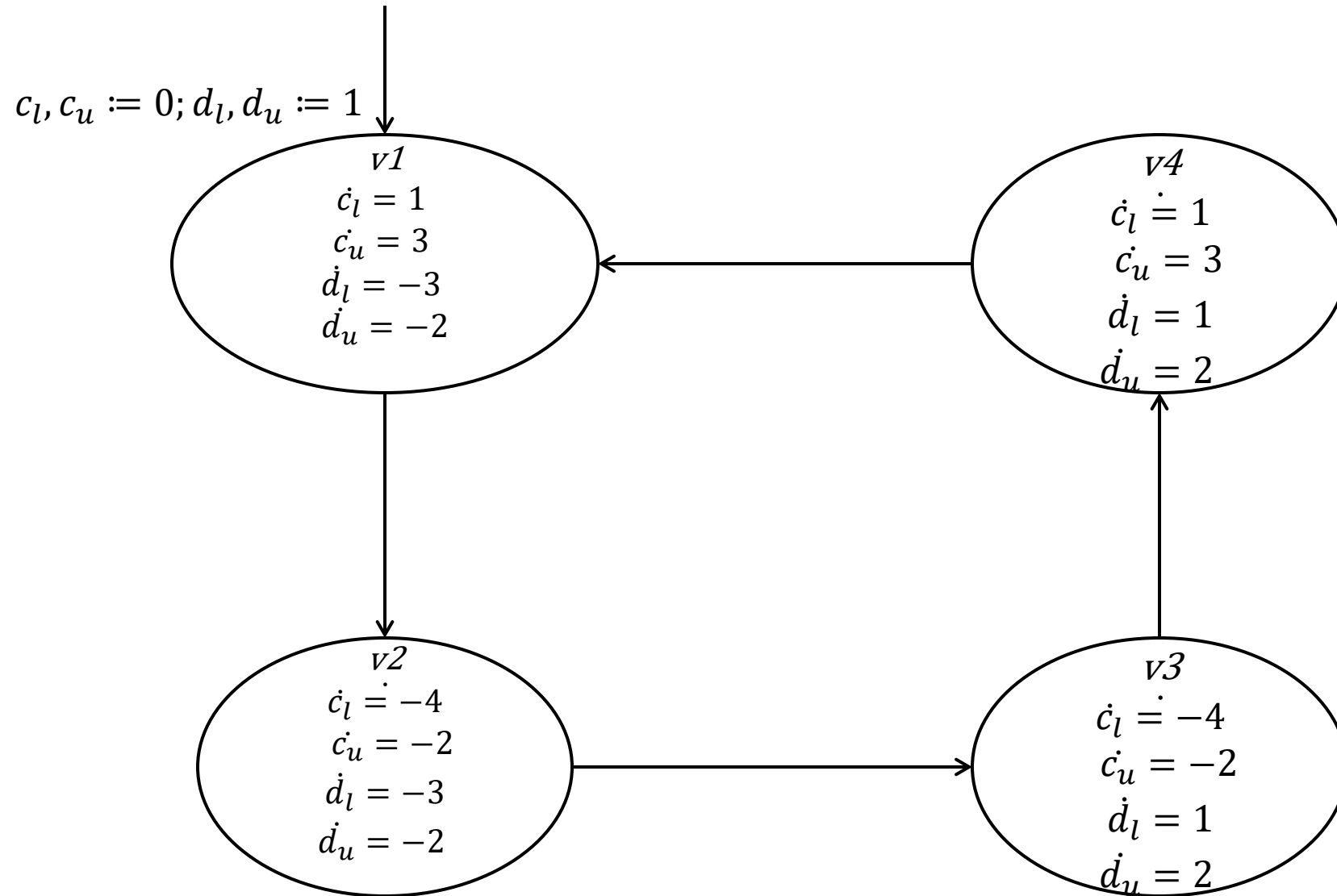
IRHA	MRA
x	$x_\ell ; x_u$
Evolve: $d(x) \in [a_1, b_1]$	Evolve: $d(x_\ell) = a_1; d(x_u) = b_1$
Eff: $x' \in [a_1, b_1]$	Eff: $x_\ell = a_1; x_u = b_1$
$x' = c$	$x_\ell = x_u = c$
Guard: $x \geq 5$	$x_l \geq 5$
	$x_l < 5 \wedge x_u \geq 5$ Eff $x_l = 5$



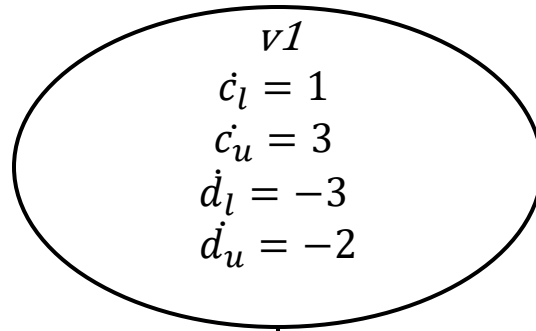
Example IRHA



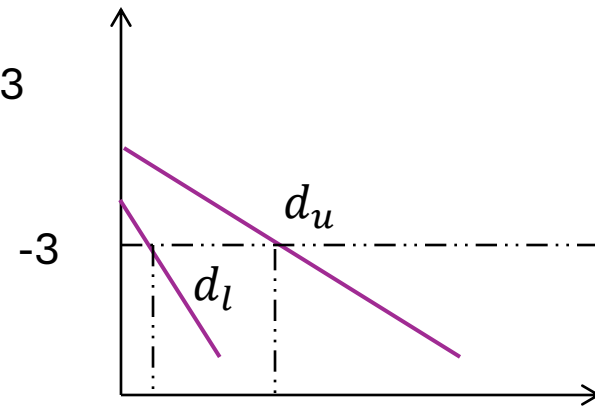
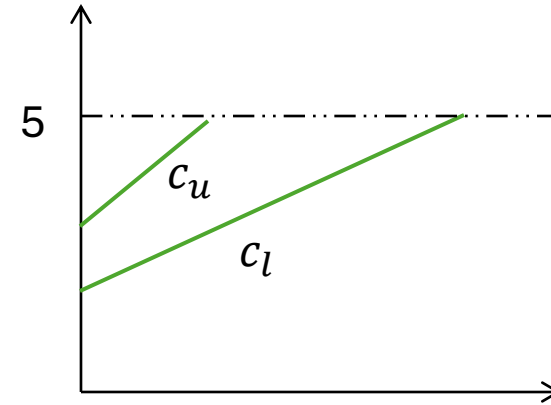
Initialized Singular HA



Transitions

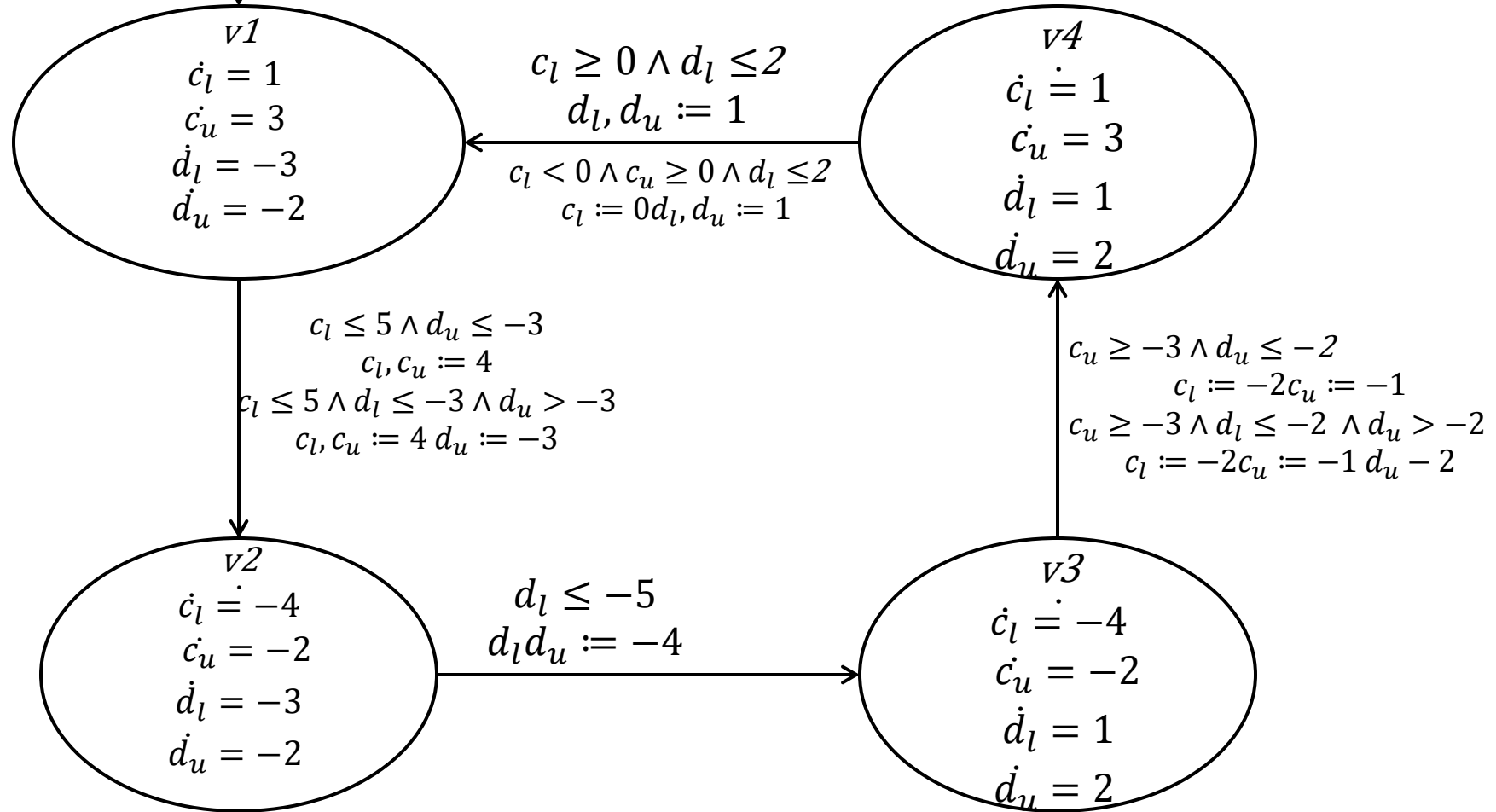


$c_l \leq 5$
 $c_l, c_u := 4$
 $d_u \leq -3$ *no reset*
 $d_u > -3 \wedge d_l \leq -3 \quad d_u := -3$



Initialized Singular HA

$c_l, c_u := 0; d_l, d_u := 1$



Can this be further generalized ?

- For initialized Rectangular HA, control state reachability is decidable
 - Can we drop the initialization restriction?
 - No, problem becomes undecidable (next time)
 - Can we drop the rectangular restriction?
 - No, problem becomes undecidable

Verification in tools

```
Algorithm: BasicReach
2 Input:  $A = \langle V, \Theta, A, D, T \rangle, d > 0$ 
    $Rt, Reach: val(V)$ 
4    $Rt := \Theta;$ 
    $Reach := \emptyset;$ 
6   While ( $Rt \not\subseteq Reach$ )
      $Reach := Reach \cup Rt;$ 
8      $Rt := Rt \cup Post_D(Rt);$ 
      $Rt := Post_{T(d)}(Rt);$ 
10 Output:  $Reach$ 
```

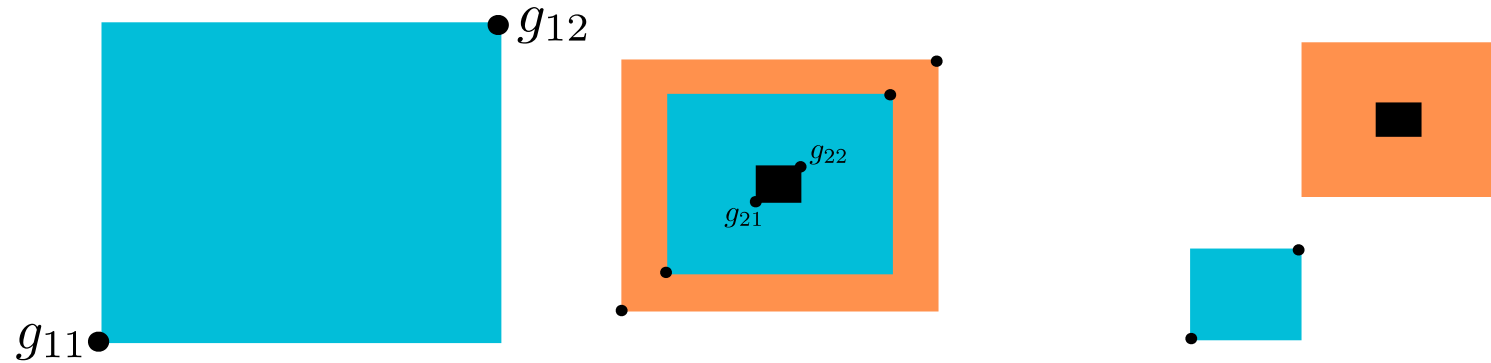
```
Algorithm:  $Post_D$ 
2  $\backslash\backslash$  computes post of all transitions
Input:  $A, D, S_{in}$ 
4    $S_{out} = \emptyset$ 
   For each  $a \in A$ 
6     For each  $\langle g_1, g_2 \rangle \in S_{in}$ 
       If  $\llbracket g_1, g_2 \rrbracket \cap \llbracket g_{a1}, g_{a2} \rrbracket \neq \emptyset$ 
8          $S_{out} := S_{out} \cup \langle g_{a1}, g_{a2} \rangle$ 
Output:  $S_{out}$ 
```

```
Algorithm:  $Post_{T(d)}$ 
 $\backslash\backslash$  computes post of all trajectories
3 Input:  $A, T, S_{in}, d$ 
    $S_{out} = \emptyset$ 
5   For each  $\ell \in L$ 
     For each  $\langle g_1, g_2 \rangle \in S_{in}$ 
7        $P := \cup_{t \leq d} \llbracket g_1, g_2 \rrbracket \oplus \llbracket t g_{\ell 1}, t g_{\ell 2} \rrbracket$ 
        $S_{out} := S_{out} \cup Approx(P)$ 
9 Output:  $S_{out}$ 
```

Data structures make reachability go around

- Hyperrectangles
 - $[g_1; g_2] = \{x \in R^n \mid \|x - g_1\|_\infty \leq \|g_2 - g_1\|_\infty\} = \Pi_i[g_{1i}, g_{2i}]$
- Polyhedra
- Zonotopes [\[Girard 2005\]](#)
- Ellipsoids [\[Kurzhan et al. 2001\]](#)
- Support functions [\[Guernic et al. 2009\]](#)
- Generalized star set [\[Duggirala and Viswanathan 2018\]](#)

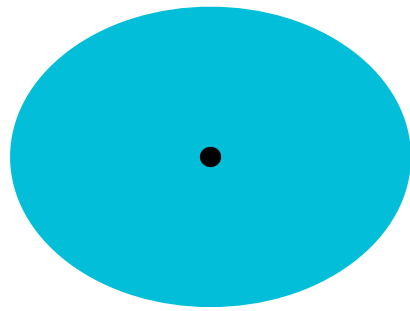
Data structures: rectangles and ellipsoids



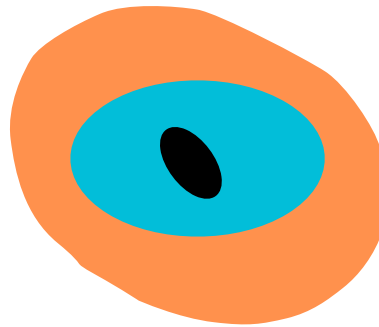
$$[[g_{11}, g_{12}]]$$

$$[[g_{11}, g_{12}]] \oplus [[g_{21}, g_{22}]] \\ = [[g_{11} + g_{12}, g_{21} + g_{22}]]$$

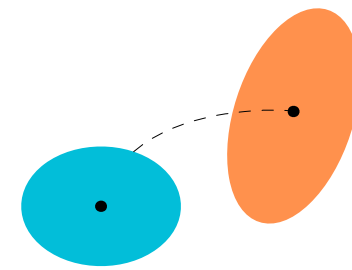
$$[[g_{11}, g_{12}]] \oplus [[t.g_1, t.g_2]]$$



$$[[c_1, Q]]$$

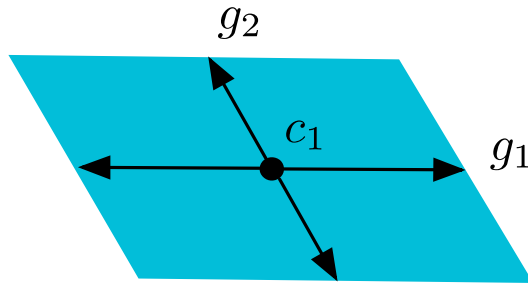


$$[[c_1, Q_1]] \oplus [[c_2, Q_2]] \neq [[c_3, Q_3]]$$

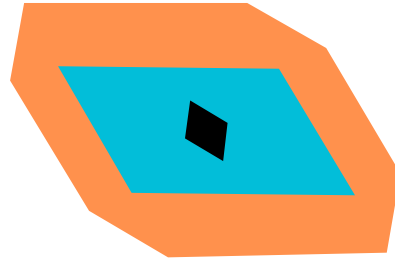


$$[[Ac_1, AQA^T]]$$

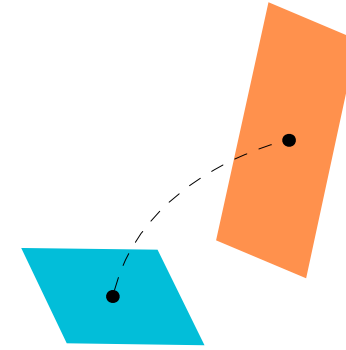
Zonotopes and polytopes



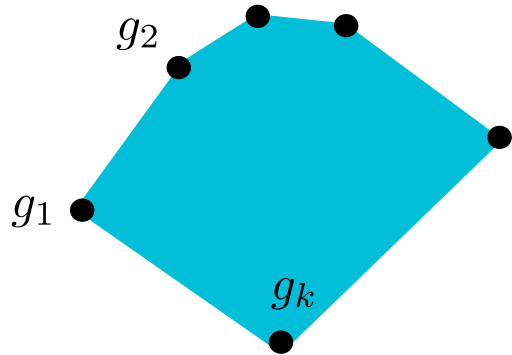
$$[[c_1, \langle g_1, g_2 \rangle]]$$



$$[[c_1, \langle g_1, g_2 \rangle]] \oplus [[c_2, \langle g'_1, g'_2 \rangle]] \\ = [[c_1 + c_2, \langle g_1, g'_1, g_2, g'_2 \rangle]]$$

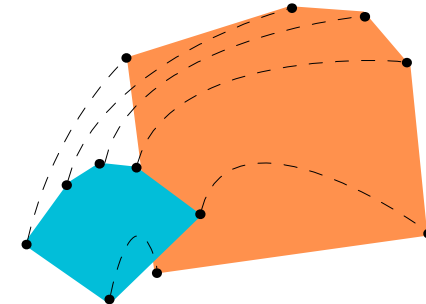
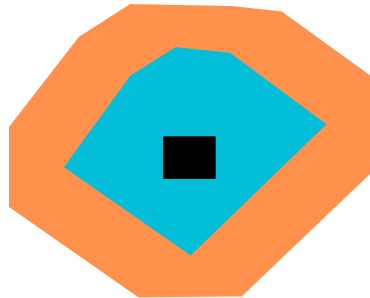


$$[[Ac_1, \langle Ag_1, Ag_2 \rangle]]$$



$$[[A, b]]$$

$$[[g_1, \dots, g_k]]$$



$$[[\xi(g_1, t), \dots, \xi(g_k, t)]]$$

Summary

- ITA: Restricted class of hybrid automata
 - Clocks, integer constraints
 - No clock comparison, linear
- Control state reachability with Alur-Dill's algorithm (region automaton construction)
- Rational coefficients; multirate Automata
- Initialized Rectangular Hybrid Automata
- HyTech, PHAVer use polyhedral reachability computations